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Method for improving plant genetic transformation and gene editing efficiency

Abstract

The present invention provides a method for improving plant genetic transformation and gene editing efficiency. Specifically, the method improves the regeneration efficiency of plant genetic transformation and/or improves the efficiency of plant gene editing by expressing genes that promote plant cell division, especially meristematic cell division.

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Background/Summary

REFERENCE TO SEQUENCE LISTING TEXT FILE

(1) The Sequence Listing text file associated with the instant disclosure has been electronically submitted to the United States Patent and Trademark Office via EFS as a 123.118 byte UTF-8-encoded text file created on Jul. 5, 2023 and entitled “1547_48_PCT_US_ST25.xml”. The Sequence Listing submitted via EFS is hereby incorporated by reference in its entirety.

TECHNICAL FIELD

(2) The present invention belongs to the field of plant genetic engineering. Specifically, the present invention relates to a method for improving plant genetic transformation and gene editing efficiency. More specifically, the present invention relates to improving regeneration efficiency of plant genetic transformation and/or improving efficiency of plant gene editing by expression of genes which can promote division of plant cells, especially meristematic cells.

BACKGROUND OF THE INVENTION

(3) Development of crop genetic breeding includes artificial selection breeding, cross breeding, mutation breeding and molecular marker assisted breeding using molecular technology. As the variety genetic diversity is gradually reduced, the bottleneck effect of traditional breeding is more and more obvious: it is difficult to use conventional breeding technology to obtain breakthrough new varieties to meet the requirement of human and sustainable agricultural development. The rapid development of life sciences makes it possible to enter the post-genome era from the “read” phase of the biological genetic information, and the accurate “rewriting” of the genome and the “new design” are becoming reality. This kind of biological technical means designed for creating new trait or living body shows great prospect in the field of disease treatment, medicine, manufacturing, especially agriculture and so on.

(4) Genome editing technology is a revolutionary technical means appearing in the current life science which can realize accurate, efficient and specific rewriting of the genome and has revolutionary pushing effect to the research and exploration of life science. Gene editing refers to deleting, replacing, or inserting operation of the target gene so as to modify the genetic information to obtain a new function or phenotype, even creating a new species. Development of efficient and accurate breeding technology suitable for crops using gene editing technology will break the defect of the traditional breeding, realizing molecular design breeding of precise transformation from the genome. It has important strategic significance for the development of future agriculture.

(5) Current gene editing technology mainly comprises ZFN, TALEN and CRISPR/Cas system. CRISPR/Cas system, due to its high efficiency and flexibility, is currently the simplest and widely used gene editing technology system. In CRISPR/Cas system, Cas protein can target any position in the genome under the guide action of the artificial designed guide RNA (guide RNA). Base editing system is a new gene editing technology developed based on CRISPR system, which can be divided into cytosine base editing system and adenine base editing system, wherein the respectively deaminase and adenine deaminase are fused with Cas9 single-chain nickase. Under the targeting function of the guide RNA, Cas9 single-chain nickase generates a single-stranded DNA region, so the deaminase can efficiently respectively remove amino group of the C and A nucleotide on the single-stranded DNA of the targeting position, resulting in U base and I base, which can be repaired as T base and G base by the repair process of the cell itself. Base editing technology overcomes the defect of traditional DSB-mediated gene editing, which can efficiently realize the precise replacement of a single base. CRISPR/Cas system-mediated robust genetic engineering technology system will provide strong technical support for gene research and new plant molecular design breeding, and will accelerate the cultivation of new variety of crops and realize sustainable development of agriculture.

(6) A key step of plant gene editing is to deliver the gene editing nuclease protein or coding nucleic acid to the plant cell to realize the editing of the target gene. The present plant genome editing delivery technology is mainly realized by genetic transformation and tissue culture technology, mainly comprising *agrobacterium* mediated method and gene gun method. Important progress has been made in plant transformation over the past few years, but transformation of many agronomically important plants (e.g., maize, soybeans, canola, wheat, indica rice, sugar cane and sorghum; and inbred lines) is still difficult and time-consuming. Generally, the only method causing the culture reaction is to optimize the culture medium components and/or explant materials and sources, which results in the success in some gene type, but many important crop plants (including excellent inbred line or variety) do not generate beneficial culture reaction and regeneration technology system. Although the transformation of a pattern genotype may be effective, the process of gradually introgressing the transgene into the product inbred line is laborious, expensive and time-consuming. Especially for monocotyledon wheat, efficiency of current gene gun and *agrobacterium* transformation method is low and greatly limited by the genotype, and a long-term tissue culture process is required. At present, the maximum bottleneck of wheat gene editing is the low efficiency of the current traditional wheat transformation system, the large technical difficulty, limitation by genotype and low throughput.

(7) In order to facilitate the research of plant gene function and molecular design breeding more efficiently utilizing gene editing technology, establishing and digging a method to improve the plant transformation efficiency and shorten the time of tissue culture has important meaning.

SUMMARY OF THE INVENTION

(8) In one aspect, the present invention provides a method for improving plant cell regeneration efficiency in plant transformation, the method comprising: (a) introducing into a cell of the plant i) an expression construct comprising a coding nucleic acid sequence of WUS, an expression construct comprising a coding nucleic acid sequence of BBM and an expression construct comprising a coding nucleic acid sequence of SERK; and/or ii) an expression construct comprising a coding nucleic acid sequence of GRF and an expression construct comprising a coding nucleic acid sequence of GIF; (b) regenerating an intact plant from the plant cell.

(9) In another aspect, the present invention provides a method for improving the transformation efficiency of an exogenous nucleic acid sequence of interest in a plant or for transforming an exogenous nucleic acid sequence of interest into a plant, the method comprising: (a) introducing into a cell of the plant i) an expression construct comprising a coding nucleic acid sequence of WUS, an expression construct comprising a coding nucleic acid sequence of BBM and an expression construct comprising a coding sequence of SERK; and/or ii) an expression construct comprising a coding nucleic acid sequence of GRF and an expression construct comprising a coding nucleic acid sequence of GIF; (b) introducing at least one expression construct comprising at least one exogenous nucleic acid sequence of interest into the plant cell; and (c) regenerating an intact plant from the plant cell.

(10) In another aspect, the present invention provides a method for improving gene editing efficiency in a plant or for gene editing in a plant, the method comprising: (a) introducing into a plant cell i) an expression construct comprising a coding sequence of WUS, an expression construct comprising a coding sequence of BBM and an expression construct comprising a coding sequence of SERK; and/or ii) an expression construct comprising a coding sequence of GRF and an expression construct comprising a coding sequence of GIF; (b) introducing at least one expression construct comprising at least one exogenous sequence of interest to the plant cell, wherein the at least one exogenous sequence of interest encodes a component of a gene editing system; or introducing at least one component of a gene editing system to the plant cell; and (c) regenerating an intact plant from the plant cell.

(11) The invention further provides a kit for carrying out the method of the invention, comprising at least i) an expression construct comprising a coding nucleic acid sequence of WUS, an expression construct comprising a coding nucleic acid sequence of BBM and an expression construct comprising a coding nucleic acid sequence of SERK; and/or ii) an expression construct comprising a coding nucleic acid sequence of GRF and an expression construct comprising a coding nucleic acid sequence of GIF.

(12) The present invention also provides use of i) an expression construct comprising a coding nucleic acid sequence of WUS; an expression construct comprising a coding nucleic acid sequence of BBM and an expression construct comprising a coding nucleic acid sequence of SERK; and/or ii) an expression construct comprising a coding nucleic acid sequence of GRF and an expression construct comprising a coding nucleic acid sequence of GIF, for improving plant cell regeneration efficiency in plant transformation, for improving the transformation efficiency of exogenous nucleic acid sequence of interest in the plant or for improving the gene editing efficiency in the plant.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1. Expression constructs for detecting the influence of a plurality of DR combinations on genetic transformation and gene editing efficiency.

(2) FIG. 2 shows expression constructs for verifying the effect of GRF4 and gif1 combination on genetic transformation and gene editing efficiency in wheat.

(3) FIG. 3 shows the expression constructs for GRF4 and gif1 combinatorial optimization experiment.

(4) FIG. 4 shows the soybean GRF/gif expression construct.

(5) FIGS. 5A-5K. The effect of GmGRF-GmGIF1 complexes on regeneration, transformation, and genome editing of the soybean cultivar Williams 82. (FIG. 5A) Schematic representations of T-DNA regions each containing one of the four GmGRF-GmGIF1 complexes. pBSE401 is the CRISPR/Cas9 control construct. (FIG. 5B) Schematic of the GmFAD2 sgRNA target sites in the two subgenomes. (FIG. 5C) The general procedure for *Agrobacterium*-mediated transformation of soybean. (FIG. 5D) Comparison of the effects of four GmGRF-GmGIF1 complexes on regeneration frequencies of Williams 82. RF (regeneration frequency)=no. of explants with multiple buds/explant number $\times$ 100%. (FIG. 5E) Numbers of putative glufosinate-resistant elongated shoots $\geq$ 2 cm in length (on day 50 after transformation) and shoots $\geq$ 9 cm in length (on day 75 after transformation) regenerated from explants transformed with the four GmGRF-GmGIF1 complexes and pBSE401, respectively. (FIG. 5F) Average numbers of elongated shoots regenerated from single explants transformed with the four GmGRF-GmGIF1 complexes and pBSE401, respectively. (FIG. 5G) Regeneration of explants transformed with pGmGRF5-GmGIF1 and pBSE401 and cultured in medium supplemented with 5.0 mg/L glufosinate on day 50 after infection. (FIG. 5H) Typical explants with regenerated shoots cultured in

medium supplemented with 2.0 mg/L glufosinate on day 75 after infection with pGmGRF5-GmGIF1 and pBSE401, respectively. (FIG. 5I) Transformation efficiencies of Williams 82 transformed with the four GmGRF-GmGIF1 complexes and pBSE401, respectively. (FIG. 5J) GmFAD2 mutation frequencies with CRISPR/Cas9 treated by four pGmGRF-GmGIF1 constructs and pBSE401, respectively. (FIG. 5K) Performance of mature GmFAD2-edited soybean plants (160 days after transformation) transformed with pGmGRF5-GmGIF1 and pBSE401, respectively. All values and error bars are mean values $\pm$ s.e.m. of three independent experiments.

(6) FIGS. 6A-6G. The effect of the GmGRF5-GmGIF1 complex on regeneration, transformation, and genome editing of two soybean cultivars Zhonghuang 13 and Hefeng 25. (FIG. 6A) Regeneration frequencies of explants transformed with pGmGRF5-GmGIF1 and pBSE401 on day 50 after transformation. (FIGS. 6B and 6C) Numbers of putative glufosinate-resistant elongated shoots (>2 cm in length) on day 50 after transformation and elongated shoots (>9 cm in length) on day 75 regenerated from explants of Zhonghuang 13 (FIG. 6B) and Hefeng 25 (FIG. 6C) transformed with pGmGRF5-GmGIF1 and pBSE401, respectively. (FIG. 6D) Shoot induction, shoot proliferation and shoot elongation in Hefeng 25 explants transformed with pGmGRF5-GmGIF1 and pBSE401, respectively. (FIG. 6E) Average numbers of elongated shoots regenerated from single explants of Zhonghuang 13 and Hefeng 25 transformed with pGmGRF5-GmGIF1 and pBSE401, respectively. (FIG. 6F) Transformation efficiencies of Zhonghuang 13 and Hefeng 25 transformed with pGmGRF5-GmGIF1 and pBSE401, respectively. (FIG. 6G) GmFAD2 mutation rates in Zhonghuang 13 and Hefeng 25 transformed with pGmGRF5-GmGIF1 and pBSE401, respectively. All values and error bars are mean values $\pm$ s.e.m. of three independent experiments.

(7) FIG. 7A-7D. The effect of the GmGRF5-GmGIF1 complex on regeneration, transformation, and genome editing efficiencies in strawberry cultivar Benihoppe. (FIG. 7A) Schematic representation of the T-DNA region containing the GmGRF5-GmGIF1 complex. pHUE411-GFP is the CRISPR/Cas9 control construct. (FIG. 7B) Effects of the GmGRF5-GmGIF1 complex on strawberry regeneration, transformation, and genome editing efficiencies. The data were collected on day 35 after explants were transformed with pHUE411-GFP-GmGRF5-GmGIF1 and pHUE411-GFP. RF (regeneration frequency)=no. of regenerated shoots/total explants $\times$ 100%. TE (transformation efficiency)=no. of transgenic shoots/total explants $\times$ 100%. MF (mutation frequency)=no. of mutants/total explants $\times$ 100%. (FIG. 7C) GFP positive calli from explants transformed with pHUE411-GFP-GmGRF5-GmGIF1 and pHUE411-GFP, on day 21 after transformation. (FIG. 7D) Calli and shoots from explants transformed with pHUE411-GFP-GmGRF5-GmGIF1 and pHUE411-GFP, respectively, on day 35 after transformation.

(8) FIGS. 8A-8H. Comparison of the effects of TaGRF4-TaGIF1 and mTaGRF4-TaGIF1 on regeneration and genome editing in two common wheat cultivars Kenong 199 and Bobwhite. (FIG. 8A) Schematic representation of common wheat GIF1, GRF4, and mutated GRF4. The dotted lines indicate the interaction between the SNH and QLQ domains. mTaGRF4 was created by introducing five point mutations in the miRNA396 target site of common wheat TaGRF4. (FIG. 8B) Schematic representations of constructs pTaGRF4-TaGIF1, pmTaGRF4-TaGIF1, and the base editor, pUBI-A3A. pUBI-GFP is the control construct. (FIG. 8C) General procedure for transgene-free genome editing in common wheat by transient expression of a cytosine base editor. (FIG. 8D) Comparison of the effects of TaGRF4-TaGIF1 and mTaGRF4-TaGIF1 on regeneration frequencies of Bobwhite and Kenong199. RF (regeneration frequency)=no. of regenerated shoots/immature embryos bombarded $\times$ 100%. (FIG. 8E) Comparison of the effects of TaGRF4-TaGIF1 and mTaGRF4-TaGIF1 on genome editing frequencies in Bobwhite and Kenong199. MF (mutation frequency)=no. of mutants/immature embryos bombarded $\times$ 100%. (FIG. 8F) Primer sets for detecting transgene-free mutants, and the outcome of tests on 22 representative TaALS mutant plants (Kenong 199). (FIG. 8G) The Transgene-free frequencies in Bobwhite and Kenong199 wheat cultivars transformed with TaGRF4-TaGIF1, mTaGRF4-TaGIF1 and pUBI-GFP (control construct), respectively. (FIG. 8H) Transgene-free mutant plants regenerated from Kenong 199 immature embryos transiently-expressing mTaGRF4-TaGIF1 and the cytosine base editor A3A-PBE, do not exhibit abnormal growth. In (FIG. 8D), (FIG. 8E) and (FIG. 8G), values and error bars are means $\pm$ s.e.m. of three independent experiments.

(9) FIGS. 9A-9C. The effect of transient expression of the mutated TaGRF4-TaGIF1 complex on common wheat regeneration and genome editing efficiencies in nine elite wheat cultivars. (FIG. 9A) The regeneration frequencies of nine elite common wheat cultivars transformed with pmTaGRF4-TaGIF1 and pUBI-GFP (control construct). Values and error bars are means $\pm$ s.e.m. of three independent experiments. (FIG. 9B) Regenerated plants of Xiaoyan 54 and Zhongmai 175 transformed with mTaGRF4-TaGIF1 and pUBI-GFP (control construct), respectively, 28 days after transformation. (FIG. 9C) The mutation frequencies in the nine elite common wheat cultivars transformed with pmTaGRF4-TaGIF1 and pUBI-GFP (control construct). MF (mutation frequency)=total no. of mutants/total immature embryos bombarded $\times$ 100%.

(10) FIGS. 10A-10C. Detection of mutations and transgene-free mutant plants in 15 representative Williams 82 soybean lines transformed with pGmGRF5-GmGIF1. (FIG. 10A) Mutations in the GmFAD2 gene from the 15 representative soybean lines identified by PCR-RE assays. Lanes 1 to 15 show digests of the PCR fragments amplified from the transgenic soybean plants using BstXI. Lanes labeled CK show the digests of PCR fragments amplified from a wild-type control plant. (FIG. 10B) The outcome of tests for transgene-free mutants using two primer sets in 15 representative gmfad2 mutant plants. Lanes without a band indicate transgene-free mutants. Lanes labeled CK are the

PCR fragments amplified from a WT plant. (FIG. 10C). Sanger sequencing of the GmFAD2 gene in wild type and the edited gmfad2 mutants.

(11) FIGS. 11A-11C. Phylogenetic analysis of GRFs from *Glycine max* (Gm) and *Fragaria vesca* (Fve) and the sgRNA designed to generate mutant strawberry FaPL genes. (FIG. 11A) Clustal W was used to align 31 GRFs, including nine FveGRFs and 22 GmGRFs. MEGA 7.0 was used to construct a neighbor-joining phylogenetic tree with 1000 bootstrap replications. (FIG. 11B) Schematic of the sgRNA designed to target the FaPL gene. (FIG. 11C) Detection of mutations by Sanger sequencing in regenerated strawberry lines transformed with pHUE411-GFP-GmGRF5-GmGIF1.

(12) FIGS. 12A and 12B. Multiple sequence alignment of the deduced GRF4 and GIF1 proteins, which are the most similar between common wheat and rice. (FIG. 12A) The GRF4 from common wheat shows 62.5% conservation with the GRF4 from rice at the amino acid level. (FIG. 12B) The GIF1 from common wheat shows 86.5% conservation with the GIF1 from rice at the amino acid level. Sequences were compared using Geneious Prime.

(13) FIG. 13. Schematic of the TaALS sgRNA target sites in the common wheat genome. SgRNA target sites within a conserved region of common wheat TaALS homeologs were targeted by the base editing systems. The EcoO109I restriction enzyme site in the sgRNA target sequence was used for mutation detection.

DETAILED DESCRIPTION OF THE INVENTION

I. Definitions

(14) In the present invention, unless otherwise specified, the scientific and technical terms used herein have the meaning commonly understood by those skilled in the art, and the protein and nucleic acid chemistry used herein, molecular biology, cell and tissue culture, microbiology, immunology related term and laboratory operation steps are widely used in the corresponding field and conventional steps. For example, the standard recombinant DNA and molecular cloning techniques used in the present invention are well known to those of skill in the art, and are more fully described in the following literature: Sambrook, J., Fritsch, E. F. and Maniatis, T., Molecular Cloning: A Laboratory Manual; Cold Spring Harbor Laboratory Press: Cold Spring Harbor, 1989 (hereinafter “Sambrook”). At the same time, for a better understanding of the present invention, the definitions and explanations of the related terms are provided below.

(15) As used herein, the term “and/or” encompasses all combinations of items connected by the term, and each combination should be regarded as individually listed herein. For example, “A and/or B” covers “A”, “A and B”, and “B”. For example, “A, B, and/or C” covers “A”, “B”, “C”, “A and B”, “A and C”, “B and C”, and “A and B and C”.

(16) When the term “comprise” is used herein to describe the sequence of a protein or nucleic acid, the protein or nucleic acid may consist of the sequence, or may have additional amino acids or nucleotide at one or both ends of the protein or nucleic acid, but still have the activity described in this invention. In addition, those skilled in the art know that the methionine encoded by the start codon at the N-terminus of the polypeptide will be retained under certain practical conditions (for example, when expressed in a specific expression system), but does not substantially affect the function of the polypeptide. Therefore, when describing the amino acid sequence of specific polypeptide in the specification and claims of the present application, although it may not include the methionine encoded by the start codon at the N-terminus, the sequence containing the methionine is also encompassed, correspondingly, its coding nucleotide sequence may also contain a start codon; vice versa.

(17) “Genome” as used herein encompasses not only chromosomal DNA present in the nucleus, but also organelle DNA present in the subcellular components (e.g., mitochondria, plastids) of the cell.

(18) The term “exogenous” with respect to sequence means a sequence that originates from a foreign species, or, if from the same species, is substantially modified from its native form in composition and/or genomic locus by deliberate human intervention.

(19) “Polynucleotide”, “nucleic acid sequence”, “nucleotide sequence”, or “nucleic acid fragment” are used interchangeably to refer to a polymer of RNA or DNA that is single- or double-stranded, optionally containing synthetic, non-natural or altered nucleotide bases. Nucleotides (usually found in their 5'-monophosphate form) are referred to by their single letter designation as follows: “A” for adenylate or deoxyadenylate (for RNA or DNA, respectively). “C” for cytidylate or deoxycytidylate, “G” for guanylate or deoxyguanylate, “U” for uridylate. “T” for deoxythymidylate, “R” for purines (A or G). “Y” for pyrimidines (C or T), “K” for G or T. “H” for A or C or T. “I” for inosine, and “N” for any nucleotide.

(20) “Polypeptide”, “peptide”, “amino acid sequence” and “protein” are used interchangeably herein to refer to a polymer of amino acid residues. The terms apply to amino acid polymers in which one or more amino acid residue is an artificial chemical analogue of a corresponding naturally occurring amino acid, as well as to naturally occurring amino acid polymers. The terms “polypeptide”, “peptide”, “amino acid sequence”, and “protein” are also inclusive of modifications including, but not limited to, glycosylation, lipid attachment, sulfation, gamma-carboxylation of glutamic acid residues, hydroxylation and ADP-ribosylation.

(21) As used herein, an “expression construct” refers to a vector suitable for expression of a nucleotide sequence of interest in an organism, such as a recombinant vector. “Expression” refers to the production of a functional product. For example, the expression of a nucleotide sequence may refer to transcription of the nucleotide sequence (such as

transcribe to produce an mRNA or a functional RNA) and/or translation of RNA into a protein precursor or a mature protein.

(22) “Expression construct” of the invention may be a linear nucleic acid fragment (including a DNA or RNA fragment), a circular plasmid, a viral vector.

(23) The “expression construct” of the present invention may comprise a regulatory sequence and a nucleic acid sequence of interest operably linked thereto. The regulatory sequence and the nucleic acid sequence of interest may be of different sources, or are of the same origin but are arranged in a manner different from that normally found in nature.

(24) “Regulatory sequence” and “regulatory element” are used interchangeably and refer to a nucleotide sequence which is located upstream (5′ non-coding sequences), within, or downstream (3′ non-coding sequences) of a coding sequence, and influence the transcription, RNA processing or stability, or translation of the associated coding sequence. The regulatory sequence may include, but is not limited to, a promoter, a translation leader sequence, an intron and a polyadenylation recognition sequence. “Promoter” refers to a nucleic acid fragment capable of controlling transcription of another nucleic acid fragment. In some embodiments of the invention, the promoter is a promoter capable of controlling gene transcription in a cell regardless of whether it is derived from the cell. The promoter may be a constitutive promoter or a tissue specific promoter or a developmentally regulated promoter or an inducible promoter.

(25) As used herein, the term “operably linked” refers to a regulatory element (e.g., but not limited to, a promoter sequence, a transcription termination sequence, etc.) and a nucleic acid sequence (e.g., a coding sequence or open reading frame), are linked such that transcription of the nucleotide sequence is controlled and regulated by the transcription regulatory element. Techniques for operably linking a regulatory element region to a nucleic acid molecule are known in the art.

(26) “Introducing” a nucleic acid molecule (e.g., an expression construct) into a plant cell refers to that the nucleic acid molecule is presented to the plant cell such that the nucleic acid molecule enter the interior of the plant cell.

(27) “Regeneration” refers to the process of growing an intact plant from one or more plant cells (e.g., plant protoplasts, callus or explants).

II. Improved Plant Transformation

(28) In one aspect, the present invention provides a method for improving plant cell regeneration efficiency in plant transformation, the method comprising: (a) introducing into a cell of the plant i) an expression construct comprising a coding nucleic acid sequence of WUS, an expression construct comprising a coding nucleic acid sequence of BBM and an expression construct comprising a coding nucleic acid sequence of SERK; and/or ii) an expression construct comprising a coding nucleic acid sequence of GRF and an expression construct comprising a coding nucleic acid sequence of GIF; (b) regenerating an intact plant from the plant cell.

(29) In another aspect, the present invention provides a method for improving transformation efficiency of an exogenous nucleic acid sequence of interest in a plant or for transforming an exogenous nucleic acid sequence into a plant, the method comprising: (a) introducing into a cell of the plant i) an expression construct comprising a coding nucleic acid sequence of WUS, an expression construct comprising a coding nucleic acid sequence of BBM and an expression construct comprising a coding sequence of SERK; and/or ii) an expression construct comprising a coding nucleic acid sequence of GRF and an expression construct comprising a coding nucleic acid sequence of GIF; (b) introducing at least one expression construct comprising at least one exogenous nucleic acid sequence of interest to the plant cell; and (c) regenerating an intact plant from the plant cell.

(30) In some embodiments of this aspect, step (a) and step (b) are carried out at the same time. In some embodiments of this aspect, step (a) is performed prior to step (b). In some embodiments of this aspect, step (b) is performed prior to step (a). In some embodiments of this aspect, step (c) is performed after step (a) and step (b).

(31) In another aspect, the present invention provides a method for improving gene editing efficiency in a plant or for gene editing in a plant, the method comprising: (a) introducing into a cell of the plant i) an expression construct comprising a coding sequence of WUS, an expression construct comprising a coding sequence of BBM and an expression construct comprising a coding sequence of SERK; and/or ii) an expression construct comprising a coding sequence of GRF and an expression construct comprising a coding sequence of GIF; (b) introducing at least one expression construct comprising at least one exogenous sequence of interest to the plant cell, wherein the at least one exogenous sequence of interest encodes a component of a gene editing system; or introducing at least one component of the gene editing system to the plant cell; and (c) regenerating an intact plant from the plant cell.

(32) In some embodiments of this aspect, step (a) and step (b) are carried out at the same time. In some embodiments of this aspect, step (a) is performed prior to step (b). In some embodiments of this aspect, step (b) is performed prior to step (a). In some embodiments of this aspect, step (c) is performed after step (a) and step (b).

(33) The invention further provides a kit for carrying out the method of the invention, comprising at least i) an expression construct comprising a coding nucleic acid sequence of WUS, an expression construct comprising a coding nucleic acid sequence of BBM and an expression construct comprising a coding nucleic acid sequence of SERK; and/or ii) an expression construct comprising a coding sequence of GRF and an expression construct comprising a

coding sequence of GIF.

(34) The present invention also provides use of i) an expression construct comprising a coding nucleic acid sequence of WUS; an expression construct comprising a coding nucleic acid sequence of BBM and an expression construct comprising a coding nucleic acid sequence of SERK; and/or ii) an expression construct comprising a coding nucleic acid sequence of GRF and an expression construct comprising a coding nucleic acid sequence of GIF, for improving plant cell regeneration efficiency in plant transformation, for improving the transformation efficiency of exogenous nucleic acid sequence of interest in the plant or for improving the gene editing efficiency in the plant.

(35) WUS (WUSCHEL), BBM (BABY BOOM) and SERK (Somatic Allogenes receptor-like kinase) are conservative development regulatory factors (DR) widely present in plants. The inventors have surprisingly found that co-expression of the combination of WUS, BBM and SERK in the plant cell can significantly improve the efficiency of the plant cell to regenerate into intact plants, and significantly improve transformation efficiency of an exogenous nucleic acid sequence of interest into the plant. When the exogenous nucleic acid sequence of interest encodes a gene editing system, the gene editing efficiency can be significantly improved.

(36) Examples of WUS, BBM and SERK suitable for use in the present invention include, but are not limited to, WUS, BBM, and SERK from *Arabidopsis*, canola, strawberry, potato, rice, tomato, soybean, corn or wheat.

(37) In some embodiments of various aspects of the invention, WUS is corn WUS (ZmWUS), BBM is the corn BBM (ZmBBM), or SERK is corn SERK (ZmSERK). In some embodiments, the ZmWUS comprises the amino acid sequence shown in SEQ ID NO: 1. In some embodiments, the ZmBBM comprises the amino acid sequence shown in SEQ ID NO: 2. In some embodiments, the ZmSERK comprises the amino acid sequence shown in SEQ ID NO: 3.

(38) In some embodiments of various aspects of the present invention, at least two or at least three or all of the coding nucleic acid sequence of WUS, the coding nucleic acid sequence of BBM, the coding nucleic acid sequence of SERK, and the at least one exogenous nucleic acid sequence of interest are placed in a same expression construct. In some embodiments, the coding nucleic acid sequence of WUS, the coding nucleic acid sequence of BBM, the coding nucleic acid sequence of SERK and the at least one exogenous nucleic acid sequence of interest are respectively placed in different expression constructs.

(39) In some embodiments of various aspects of the invention, the coding nucleic acid sequence of WUS, the coding nucleic acid sequence of BBM, and the coding nucleic acid sequence of SERK are placed in the same expression construct, while the at least one exogenous nucleic acid sequence of interest is placed in another expression construct.

(40) In some embodiments of various aspects of the present invention, the coding nucleic acid sequence of WUS, the coding nucleic acid sequence of BBM, the coding nucleic acid sequence of SERK and/or the at least one exogenous nucleic acid sequence of interest are operatively connected to transcription regulatory elements.

(41) Methods of expressing different proteins by the same expression construct are known in the art. For example, the different proteins can be placed in the same expression construct under the control of different transcriptional regulatory elements (e.g., different promoters). Alternatively, different proteins can be fused by self-cleaving peptide (e.g., 2A peptide, including but not limited to P2A, E2A; F2A and T2A, etc.), then expressed under the control of the same transcriptional control element (e.g., different promoter), so that separate different proteins can be generated by self-cleavage of the self-cleaving peptide after translation or translation. Alternatively, an internal ribosome entry site (IRES) can be inserted between the coding nucleic acid sequences of different proteins.

(42) GRFs (Growth Factors) are specific transcription factors in plants, mainly controlling plant cell size, chloroplast proliferation, stamen development, osmotic stress and other plant growth and development processes. GRF transcription factors are widely existed in the plant, mainly comprising two conserved domains: QLQ and WRC. The QLQ domain of GRFs can interact with the SNH domain (SYT N-terminal domain) in GIF (GRF-interacting factor) proteins, so as to exercise transcriptional activation function. WRC domain comprises 1 functional nuclear localization signal and 1 DNA binding motif, plays a role in DNA binding. Generally, QLQ and WRC domain are located at the N terminal of the GRFs. However, some GRFs also have a second WRC domain at the C terminal.

(43) "GIF" (GRF-interacting factor) is a protein that can form transcription co-activation factor complexes with GRF. GIFs are homologous to human transcription co-activation factor synovial sarcoma transport protein (synovial translocation protein, SYT). In *Arabidopsis thaliana*, GIF plays a role in cell proliferation during blade development and maintains proliferation ability of the meristematic cells during flower organ development.

(44) The inventors have further surprisingly found that co-expression of the combination of GRF and GIF in plant cell can significantly improve the regenerating efficiency of a plant cell into intact plant, and significantly improve the transformation efficiency of exogenous nucleic acid sequence of interest into the plant. When the exogenous nucleic acid sequence of interest encodes a gene editing system, the gene editing efficiency can be significantly improved.

(45) Examples of GRFs suitable for use in the present invention include, but are not limited to, GRF from *Arabidopsis*, canola, potato, rice, tomato, soybean, corn or wheat. Examples of GIF suitable for use in the present invention include, but are not limited to, GIF from *Arabidopsis*, canola, potato, rice, tomato, soybean, corn or wheat. However, as long as it can form transcription co-activating factor complex, GRF and GIF in the present invention does not necessarily have the same origin.

(46) In some embodiments of the invention, the GRF is wheat GRF. Suitable wheat GRFs include, but are not limited

to, wheat GRF4. In some embodiments, the wheat GRF4 comprises the amino acid sequence of SEQ ID NO: 4.

(47) In some embodiments of the invention, the GIF is wheat GIF. Suitable wheat GIFs include, but are not limited to, wheat GIF1. In some embodiments, the wheat GIF1 comprises the amino acid sequence shown in SEQ ID NO: 6.

(48) In some embodiments of the invention, the GRF is soybean GRF. Suitable soybean GRF include, but are not limited to, soybean GRF5, soybean GRF6, soybean GRF11, or soybean GRF11. In some embodiments, the soybean GRF5 comprises the amino acid sequence of SEQ ID NO: 7. In some embodiments, the soybean GRF6 comprises the amino acid sequence of SEQ ID NO: 8. In some embodiments, the soybean GRF11 comprises the amino acid sequence of SEQ ID NO: 9. In some embodiments, the soybean GRF18 comprises the amino acid sequence of SEQ ID NO: 10.

(49) In some embodiments of the invention, the GIF is derived from soybean GIF. Suitable soybean GIFs include, but are not limited to, soybean GIF1. In some embodiments, the soybean GIF1 comprises the amino acid sequence shown in SEQ ID NO: 11.

(50) Many of the transcription factors in the plant including GRF are regulated by miRNAs. For example, GRF4 is negatively controlled by miR396. The invention surprisingly found that mutating the miRNA binding site in GRF can significantly improve the effect of the GRF/GIF combination in improving efficiency of plant cell regeneration and plant genetic transformation.

(51) Therefore, in some embodiments of the present invention, the GRF comprises a mutated miRNA binding site, so as not to be regulated by the miRNA. Examples of the miRNAs include, but are not limited to, miR396, depending on the particular GRF. In some embodiments, the GRF with the mutated miRNA binding site comprises the amino acid sequence of SEQ ID NO: 5.

(52) In some embodiments of the present invention, at least two or all of the coding sequence of the GRF, the coding sequence of the GIF and the at least one exogenous nucleic acid sequence of interest are placed in a same expression construct.

(53) In some embodiments of the present invention, the coding sequence of the GRF and the coding sequence of GIF are placed in a same expression construct, while the at least one exogenous nucleic acid sequence of interest is placed in another expression construct.

(54) In some embodiments of the present invention, the coding sequence of the GRF, the coding sequence of the GIF and the at least one exogenous nucleic acid sequence are operatively linked to transcription control sequences.

(55) In some embodiments of the invention, the GRF is fused to the GIF. In some embodiments, the GRF is fused to the N terminal of the GIF. In some embodiments, the GRF is fused to the GIF through a linker. An exemplary linker comprises the sequence AAAA (SEQ ID NO: 12) or SGGS (SEQ ID NO: 13). Preferably, the linker is AAAA.

(56) In some embodiments of the present invention, “an expression construct comprising a coding nucleic acid sequence of GRF and an expression construct comprising a coding nucleic acid sequence of GIF” encompasses an expression construct comprising a coding sequence of the fusion protein of GRF and GIF.

(57) In some embodiments of the present invention, the fusion protein of the GRF and GIF comprises the amino acid sequence encoded by any one of SEQ ID NO: 17-22. In some embodiments of the present invention, the fusion protein of the GRF and GIF is encoded by any one of SEQ ID NO: 17-22. In some embodiments of the present invention, the fusion protein of GRF and GIF comprises the amino acid sequence encoded by any one of SEQ ID NO: 23-28.

(58) The “at least one exogenous nucleic acid sequence of interest” may be any nucleic acid sequence to be transformed into the plant. For example, the exogenous nucleic acid sequence of interest can encode an agronomic trait, insect resistance, disease resistance, herbicide resistance, sterility, grain characteristics, and a trait important for the commercial product. The nucleic acid sequence of interest may also include those nucleic acid sequences involved in oil, starch, carbohydrate or nutrient metabolism, and those nucleic acid sequences affecting seed size, sucrose content, and the like.

(59) In some preferred embodiments of the present invention, the at least one exogenous nucleic acid sequence encodes a component of a gene editing system, so as to carry out gene editing in the plant.

(60) “Gene editing”, also known as genome editing, uses a sequence-specific nuclease or a derivative thereof for nucleotide insertion, deletion or substitution in the genome of an organism. Gene editing generally results in site-specific double-strand break (DSB) at the desired position in the genome, and then introducing desired DNA insertion, deletion or substitution by the process of repairing DSB. However, gene editing can also cover base editing technology, transcriptional activation or inhibition, epigenetic modification technology, which does not relate to DSB, as long as it has sequence specificity.

(61) The gene editing system used is not particularly limited in the present invention. For example, a gene editing system suitable for use in the present invention includes but is not limited to zinc finger nuclease (ZFN), meganuclease (MGN); transcription activating factor-like effector nuclease (TALEN) and CRISPR (Clustered regularly interspaced short palindromic repeats) system. “Zinc finger nucleases” are artificial restriction enzymes prepared by fusing a zinc finger DNA binding domain to a DNA cleavage domain. The zinc finger DNA binding domain of a single ZFN typically contains 3-6 individual zinc finger repeats, each of which can identify a sequence of, for example, 3 bp. By combining different zinc finger repeating sequences, different genome sequences can be

targeted.

(62) Meganucleases generally refer to homing endonucleases capable of identifying a nucleic acid sequence of 14-40 base in length. Long recognition sequence allows the meganucleases to have strong specificity so as to reduce the off-target effect.

(63) “Transcription activator-like effector nucleases” are restriction enzymes that can be engineered to cleave specific DNA sequences, and that are typically prepared by fusing the DNA-binding domain of a transcription activator-like effector (TALE) to a DNA cleavage domain. TALE can be engineered to bind almost any desired DNA sequence.

(64) “CRISPR system” generally comprises two components capable of forming complexes with sequence specificity: CRISPR nuclease or a variant thereof, and a corresponding guide RNA. Therefore, for the CRISPR system, the at least one exogenous nucleic acid sequence of interest of the invention can comprise a nucleic acid sequence encoding a CRISPR nuclease or a variant thereof, and/or a coding nucleic acid sequence of a corresponding guide RNA.

Alternatively, at least one component of the gene editing system introduced into the plant cell may include a CRISPR nuclease or a variant thereof, and/or a corresponding guide RNA.

(65) In some preferred embodiments of the present invention, the gene editing system is a CRISPR system. A large number of different CRISPR gene editing systems are known in the art, which can be applied to the invention. For example, suitable CRISPR gene editing system can be found at the Addgene company website (Addgene, Watertown, Massachusetts). CRISPR gene editing systems cover the systems capable of changing genomic sequence, but also comprise the systems for transcription control but not changing the genomic sequence.

(66) As used herein, the term “CRISPR nuclease” generally refers to a nuclease present in the naturally occurring CRISPR system. The CRISPR nuclease variant comprises a modified form of natural CRISPR nuclease, artificial mutant (including the nickase mutant), catalytic active fragment, or a fusion with other functional protein/polypeptide and so on. A variety of artificial functional variants of CRISPR nuclease are known in the art, such as high specific variant or nickase variant, or a cytidine deaminase or adenosine deaminase fusion protein and so on. CRISPR nuclease or a variant thereof can interact with the corresponding guide RNA for recognizing, binding and/or cutting target nucleic acid structure. Those skilled in the art know how to select a suitable CRISPR nuclease or a variant thereof to achieve the purpose of the present invention.

(67) CRISPR nuclease or a variant thereof used in the CRISPR gene editing system of the invention can be selected from Cas3, Cas8a, Cas5, Cas8b, Cas8c, Cas10d, Cse1, Cse2, Csy1, Csy2, Csy3; GSU0054, Cas10, Csm2, Cmr5, Cas10, Csx11, Csx10, Csf1, Cas9, Csn2, Cas4, Cpf1 (Cas12a), C2c1, C2c3 or C2c2 protein, or functional variants of these nucleases.

(68) In some embodiments of the present invention, the CRISPR nuclease or a variant thereof comprises a Cas9 nuclease or a variant thereof. CRISPR gene editing system based on Cas9 nuclease or variant thereof is also referred to herein as CRISPR-Cas9 gene editing system. The Cas9 nuclease can be a Cas9 nuclease from different species, such as spCas9 (having an amino acid sequence of SEQ ID NO: 15) from *Streptococcus pyogenes* (*S. pyogenes*).

(69) Cas9 nuclease variant can include Cas9 nickase (nCas9), wherein one of two sub-domain (HNH nucleic acid enzyme sub-domain and RuvC sub-domain) of the Cas9 nuclease DNA cutting domain is inactivated to form a nickase. In some embodiments, combination of a Cas9 nickase and two gRNAs targeting upstream and downstream of the sequence to be edited can be used to realize deletion of the sequence to be edited, or to realize the replacement of the sequence to be edited in the presence of a donor sequence.

(70) In some embodiments of the present invention, the CRISPR nuclease or variant thereof may also comprises Cpf1 (Cas12a) nuclease or a variant thereof such as a high specific variant. The Cpf1 nuclease can be Cpf1 nuclease from different species, such as from *Francisella novicida* U112; *Acidaminococcus* sp.BV3L6 and *Lachnospiraceae* bacterium ND2006. CRISPR gene editing system based on Cpf1 nuclease or variant thereof is also referred to herein as CRISPR-Cpf1 system.

(71) In some embodiments of the present invention, the CRISPR nuclease variant further comprises a base editor. The base editor is typically a fusion protein comprising a deaminase and a CRISPR nuclease variant lack of DNA cleavage activity.

(72) As used in the present invention, the CRISPR nuclease variant lack of DNA cleavage activity comprises but not limited to Cas9 nickase (nCas9), nuclease-dead Cas9 nuclease (dCas9) or nuclease-dead Cpf1 nuclease (dCpf1). Nuclease-dead Cas9 nuclease (dCas9) or nuclease-dead Cpf1 nuclease (dCpf1) completely lacks DNA cutting activity. A plurality of CRISPR nuclease variants lack of DNA cleavage activity are known in the art.

(73) As used in the present invention, “deaminase” refers to an enzyme that catalyzes the deamination reaction. In some embodiments of the present invention, the deaminase is a cytosine deaminase, which is capable of receiving single-stranded DNA as a substrate and capable of catalyzing cytidine or deoxycytidine respectively deaminated as uracil or deoxyuracil. In some embodiments of the present invention, the deaminase is adenosine deaminase, which is capable of receiving single-stranded DNA as a substrate and capable of catalyzing adenosine or deoxyadenosine (A) to form inosine (I). A variety of suitable cytosine deaminases or adenine deaminases with single-stranded DNA as substrate are known in the art. Suitable cytosine deaminases include, but are not limited to, APOBEC1 deaminase, activation-induced cytidine deaminase (AID), APOBEC3G, CDA1, human APOBEC3A deaminase. In some preferred

embodiments, the cytosine deaminase, the cytosine deaminase is human APOBEC3A. Examples of suitable adenine deaminases include, but are not limited to, the DNA-dependent adenine deaminase disclosed by Nicloe M. Gaudelli et al. (doi: 10.1038/nature24644, 2017).

(74) By using a fusion of a CRISPR nuclease variant lack of DNA cleavage activity and a deaminase (forming a so-called “base editor”), base editing in the target nucleotide sequence, such as conversion from C to T or conversion from A to G, can be achieved. A variety of base editors are known in the art, and those skilled in the art will know how to select a suitable base editor to achieve the object of the present invention. The base editor-based CRISPR gene editing system is also referred to as a base editing system.

(75) In some preferred embodiments of the present invention, the CRISPR system is a base editing system. Preferably, the base editing system comprises a base editor having the amino acid sequence shown in SEQ ID NO: 14.

(76) As used herein. “guide RNA” and “gRNA” can be interchangeably used, which refers to a RNA molecule that can form a complex with the CRISPR nuclease or its functional variant and is capable of targeting the complex to a target sequence because it has a certain identity to the target sequence. The guide RNA targets the target sequence through base pairing between the guide RNA and the complementary strand of the target sequence. For example, gRNA used by Cas9 nuclease or its functional mutant is often composed of crRNA and tracrRNA molecules that are partially complemented to form the complex, wherein crRNA contains a guide sequence (referred to as seed sequence) that has sufficient identity to the target sequence so as to be hybridized with the complementary strand of the target sequence and directs a CRISPR complex (Cas9+crRNA+tracrRNA) to specifically bind to the target sequence. However, it has been known in the art that single guide RNA (sgRNA) can be designed, which simultaneously contains the features of crRNA and tracrRNA, gRNA used by Cpf1 nuclease or its functional variant is often only composed of matured crRNA molecules, which is also referred to as sgRNA. Designing suitable gRNA based on the CRISPR effector protein as used and the target sequence to be edited is within the skill of those skilled person in the art.

(77) In some specific embodiments of the invention, the guide RNA is a sgRNA. For example, the sgRNA comprises a scaffold shown in SEQ ID NO: 16.

(78) The sequence specific nuclease for gene editing in the invention, such as zinc finger nuclease, transcription activating factor-like effector nuclease or CRISPR nuclease or a variant thereof, may further comprise a sub-cellular localization signal (such as a nuclear localization signal), a peptide linker, a detectable label and other elements. For example, the base editor in the CRISPR base editing system generally comprises one or more nuclear localization signal (NLS) for entering the cell nucleus to realize the editing of the chromosomal DNA.

(79) The expression construct of the invention can be introduced into the plant cell by a variety of methods known in the art, the methods comprise but are not limited to gene gun method. PEG-mediated protoplast transformation and *Agrobacterium*-mediated transformation.

(80) In some embodiments of the present invention, the plant cell of the invention is a cell suitable for regenerating an intact plant cell by tissue culture. Examples of suitable plant cells include, but are not limited to, protoplast cells, callus cells, immature embryonic cells, and explants cells.

(81) Methods for regenerating a transformed intact plant by culturing the transformed protoplast, callus, immature embryo or explant is known in the art. In the regeneration process, the transformant can also be selected based on the selectable marker carried on the introduced expression construct. In some embodiments, the regeneration is carried out in the absence of a selection pressure. In some embodiments, the transformant can be selected under a moderately stringent selection condition. The moderately stringent condition refers to a condition that does not completely inhibit the growth of the non-transformants. For example, moderately stringent selection does not inhibit the growth of the transformants but partially inhibit the growth of the non-transformants. For example, under moderately stringent selection, the non-transformants can grow but slower or weaker than the transformant. The moderately stringent selection can be determined by those skilled in the art for specific plants and specific selectable markers.

(82) In some embodiments of the invention, the expression construct of the invention is transiently transformed into the plant cell. Transient transformation refers to introducing the construct into the cell, allowing it to exert the function but not integrated into the cell genome. This is particularly useful for gene editing, because transgene-free modified plants can be produced. Another surprising discovery of the present invention is that even transient expression of the combination of WUS, BBM and SERK, or a combination of GRF and GIF, can improve the efficiency of regeneration, transformation and/or gene editing of the plant.

(83) Plants suitable for transformation or gene editing using the methods of the present invention may be monocotyledonous plants or dicotyledonous plants. For example, examples of the plants include, but are not limited to, wheat, strawberry, rice, corn, soybean, sunflower, sorghum, canola, alfalfa, cotton, barley, millet, sugar cane, tomato, tobacco, cassava and potato.

(84) The method of the invention is particularly suitable for genetic transformation or gene editing in a plant variety or genotype that previously is difficult to be transformed. In some specific embodiments, the plant is wheat, for example, the wheat is Jimai 20, Jimai 22, Beijing 411, Shannong 20, Shannon 116, Xiaoyan 54, Zhongmai 27, Zhongmai 28 and Zhongmai 175. In some specific embodiments, the plant is soybean, for example, the soybean is Williams 82, zhonghuang 13 and Hefeng 25. In some specific embodiments, the plant is strawberry, such as strawberry Benihoppe.

(85) In order to obtain effective expression in the plant, in some embodiments of the invention, the coding nucleic acid sequence or the nucleic acid sequence of interest is codon optimized against the plant species of which the genome is to be modified.

(86) Codon optimization refers to a process of modifying a nucleic acid sequence for enhanced expression in the host cells of interest by replacing at least one codon (e.g. about or more than about 1, 2, 3, 4, 5, 10, 15, 20, 25, 50, or more codons) of the native sequence with codons that are more frequently or most frequently used in the genes of that host cell while maintaining the native amino acid sequence. Various species exhibit particular bias for certain codons of a particular amino acid. Codon bias (differences in codon usage between organisms) often correlates with the efficiency of translation of messenger RNA (mRNA), which is in turn believed to be dependent on, among other things, the properties of the codons being translated and the availability of particular transfer RNA (tRNA) molecules. The predominance of selected tRNAs in a cell is generally a reflection of the codons used most frequently in peptide synthesis. Accordingly, genes can be tailored for optimal gene expression in a given organism based on codon optimization. Codon usage tables are readily available, for example, at the “Codon Usage Database” available at <<www>>.<<kazusa>>.<<or>><<jp>><</codon/>> and these tables can be adapted in a number of ways. See Nakamura, Y., et al. “Codon usage tabulated from the international DNA sequence databases: status for the year 2000” Nucl. Acids Res. 28:292 (2000).

(87) In one aspect, the invention provides a plant obtained by the method of the invention and progeny thereof.

EXAMPLES

(88) Further understanding of the present invention can be obtained by reference to some specific examples given herein, which examples are only used to illustrate the invention, and are not intended to limit the scope of the invention. Obviously, various modifications and changes may be made to the present invention without departing from the spirit of the present invention, and therefore these changes and variations are also within the scope of the present application.

Example 1. Development Regulatory Factor WUS, SERK, LEC1, KN1, BBM Improving Wheat Genetic Transformation and Gene Editing Efficiency

(89) Recently, using the totipotency of plant cell, ectopic expression of specific DR combination (such as WUS, STM, MP, BBM and so on) in somatic cell has the potential for inducing meristematic tissues. Co-expression of DRs and gene editing system can greatly shorten the plant tissue culture process to obtain the genome edited plant individuals. In addition, researches showed that by expressing some genes associated with the cell division of meristematic tissues can also achieve plant genome editing without tissue culture. The development of new plant cell (especially apical meristematic tissue cells) division-promoting and development-related genes, is of important significance especially for plant such as wheat which is difficult to transform.

(90) This example is to select candidate genes which may promote plant cell division and plant regeneration. These candidate genes include WUS, BBM, SERK, KN1 and LEC.

(91) 1.1. Vector Construction

(92) Corn WUS, SERK, LEC, BBM and KN1 were selected to test. The following plant expression vectors were constructed: 1. CRISPR/Cas9 knockout vector: UBI-Cas9 2. The single base editing vector. UBI-A3A 3. sgRNA expression vector: TaU6-sgRNA 4. ZmWUS/BBM combination: nos-ZmWUS-ubi-ZmBBM 5. ZmWUS/BBM/SERK combination: nos-ZmWUS-ubi-ZmBBM-ZmSERK 6. ZmWUS/BBM/LEC1 combination: nos-ZmWUS-ubi-ZmBBM-ZmLEC1 7. ZmWUS/KN1/LEC combination: nos-ZmWUS-ubi-ZmKN1-ZmLEC1

(93) The structures of the plant expression vectors are shown in FIG. 1. The vectors 4-7 were also referred to as promoting (booster) vectors.

(94) 1.2 Transforming Wheat Immature Embryo with Gene Gun and Analyzing Transformation and Editing Efficiency

(95) UBI-Cas9 or UBI-A3A respectively combined with TaU6-sgRNA and four promoting (booster) expression vectors were co-transformed. Wheat ALS gene was selected as the editing site. The established wheat gene gun transformation technology was used to transform. Plants were obtained after tissue culture and regeneration, and mutants were detected by PCR/RE method. The results of UBI-A3A are shown in Table I below.

(96) TABLE-US-00001 TABLE 1 No. of Selective No. of booster plasmid bombardment agent mutants nos-ZmWUS-ubi- 5 F7.5 10 ZmBBM Ubi-A3A 5 F7.5 9 UB1-A3A 5 F7.5 4 nos-ZmWUS-ubi- 5 F7.5 17 ZmBBM-ZmSERK nos-ZmWUS-ubi- 5 F7.5 4 ZmBBM-ZmLEC1 nos-ZmWUS-ubi- 5 F7.5 7 ZmKN1-ZmLEC1

(97) The above results showed that: compared with the control UBI-A3A without booster, the efficiency of nos-ZmWUS-ubi-ZmBBM-ZmSERK has significant improving effect for base editing system UBI-A3A.

Example 2. Plant Growth Control Factor GRF/Gif Improve Wheat Genetic Transformation and Gene Editing Efficiency

(98) GRF transcription factors are specific transcription factors in plants, mainly control plant cell size, chloroplast proliferation, gynogenesis, osmotic stress and other plant growth and development processes. GRF and GIF can form transcription co-activation factor complex. GIFs are homologous to human transcription co-activation factor synovial sarcoma transport protein (synovial translocation protein. SYT). In plants, the GRF4 and gif1 plays a role in cell proliferation and maintains the proliferation ability of the meristematic cell during the flower organ development. The

inventors selected wheat GRF4 and wheat gif to detect whether the genetic transformation and gene editing of the wheat can be improved.

(99) 2.1. Vector Construction

(100) Wheat GRF4 and gif1 were fused by linker AAAA; the ZmUBI promoter was used for driving expression. In addition, to reduce the number of vectors, the GRF4-AAAA-gif1 fusion protein and Cas9 or A3A-nCas9 were constructed on the same plant expression vector, and the P2A was used for separate expression. The structure of the constructed vector is shown in FIG. 2.

(101) 2.2. Transforming Wheat Immature Embryo with Gene Gun and Analyzing Transformation and Editing Efficiency

(102) The established wheat gene gun transformation technology was used to co-transform the combination of the booster vector and TaU6-sgRNA constructed in 2.1; combination of UBI-Cas9 or UBI-A3A and TaU6-sgRNA was co-transformed as a control. Wheat ALS gene was selected as the editing site. Plants were obtained after tissue culture and regeneration, and mutants were detected by PCR/RE method. The results are shown in the following table.

(103) TABLE-US-00002 TABLE 2 No. of No. of No. of Plasmid bombarded bombardment seedling mutants Ubi-GRF4-gif1-Cas9 5 158 12 Ubi-Cas9 5 91 0 UBI-GRF4-gif1-A3A 5 F7.5 21 Ubi-A3A 5 F7.5 6

(104) The above results show that: as compared with the control UBI-Cas9 and UBI-A3A, UBI-GRF4-gif1-Cas9 and UBI-GRF4-gif1-A3A with wheat GRF4/gif1 added showed significantly increased number of mutants, which was increased by a factor of about 3-12. The plant growth control factors can be used as important regulatory proteins to promote plant genetic transformation and improve efficiency of genome editing.

Example 3. Optimizing GRF4/Gif1 to Improve the Wheat Gene Editing Efficiency

(105) In order to further improve the efficiency of wheat genome editing, the linker of the GRF4 and gif1 fusion protein was optimized. The linkerA (AAAA) was replaced by linkerS (SGGS). In addition, the miR396/GRF control pattern is very conservative in plants, miR396 has down-regulation effect to GRF4. The miR396 binding site of GRF4 was mutated, which may increase expression of GRF4, so as to improve the efficiency of wheat genome editing. The booster vectors shown in FIG. 3 were constructed.

(106) The vectors respectively combined with the wheat ALS gene targeting A3A base editing system in the above Examples were transformed through gene gun into wheat immature embryo. The results shown in the following table.

(107) TABLE-US-00003 TABLE 3 Transforma- No. of No. of No. of Booster plasmid tion time bombardment seedling mutants linkerA-GRF4-gif1 2019 Sep. 24 3 275 71 linkerA-mGRF4-gif1 2019 Sep. 24 3 403 125 linkerS-GRF4-gif1 2019 Sep. 24 2 154 12 linkerS-mGRF4-gif1 2019 Sep. 24 2 182 28 P2A-GRF4-gif1 2019 Sep. 26 4 191 4 P2A-mGRF4-gif1 2019 Sep. 26 3 165 1

(108) As can be seen from the above results, editing efficiency of linkerA was higher than LinkerS. Most importantly, the mutation of the binding site of the miR396 of GRF4 increased the regeneration efficiency of the wheat with large amplitude, and also increased the mutation efficiency in large amplitude.

Example 4. GRF4/Gif1 Improves the Editing Efficiency in Wheat of Different Genotype

(109) In order to investigate whether GRF4/gif1 can improve the editing efficiency of other wheat varieties, namely breaking the limitation of wheat genotype on genetic transformation, several varieties with low transformation efficiency by traditional methods were selected for testing. The results are as follows:

(110) TABLE-US-00004 TABLE 4 No. of Booster Transforma- bombard- Selective No. of No. plasmid tion time ment agent mutants Xiaoyan 54 Pm20.25 linkerA- 2019 May 16 5 F7.5 12 GRF4-gif1 Pm20.27 / 2019 May 16 5 F7.5 0 Jimai 22 Pm20.26 linkerA- 2019 May 16 5 F7.5 1 GRF4-gif1 Pm20.28 / 2019 May 16 5 F7.5 0

(111) The results show that the wheat growth transcription factor GFR4/gif1 can significantly realize editing of a plurality of varieties, breaking the restriction of genotypes.

Example 5. GRF4/Gif1 Improves Wheat Transformation Efficiency and Gene Editing Efficiency Mediated by *agrobacterium*

(112) In order to further investigate whether GRF4/gif1 can improve the efficiency of wheat transformation efficiency and gene editing mediated by *agrobacterium*, the GRF4/gif1 was constructed on the *agrobacterium* gene editing vector, transfecting wheat immature embryo by *agrobacterium*. The results are shown in the following table.

(113) TABLE-US-00005 TABLE 5 Recipient No. of young transformed *agrobacterium* material embryo mutant ubi-GRF4-A3A-ALS174-Hyg Kenong 199 519 12 A3A-ALS174-Hyg Kenong 199 199 0

(114) It can be seen that GRF/gif can significantly improve the editing efficiency mediated by *agrobacterium* transformation, increasing the gene editing efficiency from zero to 2.3%.

Example 6. GRF/Gif Improves Soybean Transformation Efficiency and Gene Editing Efficiency Mediated by *Agrobacterium*

(115) 6.1. Soybean GRF/Gif Vector Construction

(116) The result of the above Examples shows that the plant growth factor GRF/gif combination can improve the regeneration efficiency of tissue culture of monocotyledon wheat and efficiency of gene editing. In order to investigate whether GRF can increase tissue culture plant regeneration efficiency of dicotyledonous plants, endogenous GRF genes of dicotyledon soybean were selected to test whether they can improve the plant regeneration efficiency of

soybean. GFR5, GRF6, GRF11, GRF18 of soybean were selected for test, and connected with gif1 of soybean by AAAA linker. The fusion proteins and Cas9 were separated and linked by P2A; the expression was driven by 35s promoter; the sgRNA was initiated by the U6-26 promoter; and the sgRNA was constructed on the same carrier. The constructed expression vector is shown in FIG. 4.

(117) 6.2 Soybean GRF/Gif Improve the Regeneration Efficiency of Soybean Callus

(118) The above constructed vectors were transformed into soybean by *agrobacterium*; the result of the soybean regeneration efficiency is shown in the following table.

(119) TABLE-US-00006 TABLE 6 20200303 elongated elongation project starting time RECO SBA EBA ETC seedlings/explants rate 35S-GmGFR5-AAAA- 187 147 178 144 35/124 0.282258 Gmgif1 35S-GmGFR6-AAAA- 195 100 64 56 3/56 0.053571 Gmgif1 35S-GmGFR18-AAAA- 188 161 156 124 9/124 0.072581 Gmgif1 35S-GmGFR11-AAAA- 194 168 160 157 14/124 0.112903 Gmgif1 pKSE411-shRNA 178 107 72 64 4/60 0.066667

(120) RECO: representing soybean infection, recovery culture stage; SBA: represents the soybean entering selection stage; EBA: represents the soybean enters the elongation stage. ETC: represents soybean elongation culture stage, not selection added.

(121) The result shows that the soybean endogenous GFR5 in combination with gif1 can significantly improve the regeneration efficiency of soybean callus. In addition, the gene editing efficiency of the endogenous gene of the regeneration plant is detected, showing that the gene editing efficiency was also significantly improved.

Example 7. The GmGRF5-GmGIF1 Boosts Soybean Regeneration

(122) Of the 22 GRF genes predicted in the soybean genome, GmGRF5, GmGRF6, GmGRF11, and GmGRF18, which are specifically (or preferentially) expressed in flower and shoot apical meristems (Chen et al., 2019), were chosen to form fusion proteins with soybean GmGIF1, whose homolog gene in *Arabidopsis* and rice strongly expresses in flowers and shoot apical meristems to control plant growth (Kim, 2019). Each GmGRF-GmGIF1 fusion protein was co-expressed with a CRISPR/Cas9 expression cassette, generating four constructs (pGmGRF5-GmGIF1, pGmGRF6-GmGIF1, pGmGRF11-GmGIF1, and pGmGRF18-GmGIF1) (FIG. 5A), targeting two copies of the soybean fatty acid desaturase 2 (GmFAD2) gene (FIG. 5B), whose product catalyzes the conversion of oleic acid to linoleic acid and lowers the quality of soybean oil quality (Haun et al., 2014). The transformation efficiencies and frequencies of editing of GmFAD2 by the GmGRF-GmGIF1 complexes were evaluated.

(123) As shown in FIG. 5C, each of the four pGmGRF-GmGIF1 constructs was transformed into soybean cultivar Williams 82, one of the most popular transformable genotypes, by *Agrobacterium*-mediated transformation (Jia et al., 2015). Construct pBSE401 (Xing et al., 2014), which contains the GmFAD2-targeting CRISPR/Cas9 expression cassette but no GmGRF-GmGIF1 complex, was used as control (FIG. 5A). The number of explants recovered after introduction of each of the four GmGRF-GmGIF1-containing constructs was similar to that obtained with pBSE401 (Tables 7 and 8). However, after two weeks of bud induction and two weeks of shoot proliferation with glufosinate selection, the explants transformed with pGmGRF5-GmGIF1 produced large numbers of buds: 93.8% of the explants in each transformation event produced multiple buds, compared with 58.4% in the control and similar proportions in the explants infected with other GmGRF-GIF1 constructs (FIG. 5D, Tables 7 and 8). On elongation medium with moderate stringency of glufosinate selection, the average number of elongated shoots (≥ 2 cm) increased 2.8-fold with pGmGRF5-GmGIF1 (432.3) relative to the control (156.0) and more than 2.5-fold relative to the other constructs (FIGS. 5E and 1G, Tables 7 and 8). Similarly, average numbers of elongated large shoots (≥ 9 cm) were 2.5-fold greater with pGmGRF5-GmGIF1 (85.3) than with pBSE401 (33.7) (FIGS. 5E and 5H, Tables 7 and 8). The inventors also found that the explants transformed with pGmGRF5-GmGIF1 produced significantly more putative glufosinate resistant shoots per explant (3.0) than the control (1.1) (FIGS. 5F, 5G and 5H, Tables 7 and 8). Also, the time from the initial transformation to when the first elongated shoot reached 9 cm was 58 days for pGmGRF5-GmGIF1, compared with 70 for the control (Tables 7 and 8). There were no significant differences in shoot induction, shoot elongation, and time from the initial transformation to first 9 cm shoot between explants infected with pBSE401 and the three other GmGRF-GmGIF1 complexes (Tables 7 and 8). Collectively, these data show that the GmGRF5-GmGIF1 complex stimulates regeneration of soybean explants.

Example 8. The GmGRF5-GmGIF1 Complex Enhances Genome Editing in Soybean

(124) Efficient regeneration is a prerequisite for successful plant genetic transformation and genome editing. Consistent with the above regeneration frequencies, the transformation efficiency (TE) of explants when pGmGRF5-GmGIF1 was used was 21.8%, significantly higher than the 8.5% with pBSE401, while the other complexes all yielded frequencies of no more than 10.2% (FIG. 5I, Tables 7 and 8). The inventors used PCR-RE assays and Sanger sequencing to detect mutations of GmFAD2 in the regenerated lines. Again, in agreement with the TEs, explants transformed with pGmGRF5-GmGIF1 contained more edits and significantly higher mutation frequencies (MF) than those transformed with pBSE401 (FIG. 5J, Tables 7 and 8). Also, a large number of the transformants had mutations in both FAD2-1A and FAD2-1B, indicating that both genes were efficiently mutated (FIG. 10, Table 9). Intriguingly, among the 94 mutants from transformation with pGmGRF5-GmGIF1, 15 (16.0%) were transgene-free. (FIG. 10, Tables 7 and 8).

(125) To see whether the constitutive expression of GmGRF-GmGIF1 complexes caused phenotypic abnormalities,

the inventors grew GmFAD2-edited Williams 82 plants transformed with pGmGRF5-GmGIF1 and pBSE401 in a greenhouse. Throughout the whole growth period, Williams 82 plants transformed with pGmGRF5-GmGIF1 were fertile and no morphologic differences were observed between the two sets of lines (FIG. 5K), suggesting that that GmGRF5-GmGIF1 greatly stimulates genome-editing of soybean without producing undesirable side-effects. Example 9. The GmGRF5-GmGIF1 Complex Enables Genome Editing of Marginally Transformable Soybean Cultivars

(126) Because the pGmGRF5-GmGIF1 construct yielded the highest rates of regeneration, genetics transformation, and mutation among the four GmGRF-GmGIF1 complexes, it was used in an attempt to increase the transformation and genome-editing efficiencies in two marginally transformable but commercially important soybean cultivars, Zhonghuang 13 and Hefeng 25. The GmGRF5-GmGIF1 complex was transformed into Zhonghuang 13 and Hefeng 25 by *Agrobacterium*-mediated transformation, in combination with a GmFAD2-targeting CRISPR/Cas9 expression cassette, with pBSE401 as control. In Zhonghuang 13, GmGRF5-GmGIF1 gave rise to substantially higher frequencies of regeneration (84.0%), elongated putative glufosinate-resistant shoots (312.6, ≥ 2 cm in length), and elongated putative glufosinate-resistant shoots per explant (2.6), as well as higher TE (18.2%) and MF (16.0%), than did pBSE401 (65.6%, 233.0, 1.7, 5.3% and 2.4%, respectively) (FIGS. 6A, 6B, 6E and 6G, Tables 7 and 8). In Hefeng 25, about 71.1% of explants transformed with pGmGRF5-GmGIF1 produced at least one elongated shoot ≥ 2 cm, whereas none of the explants transformed with pBSE401 produced shoots (FIG. 6A, 6C-6E, Tables 7 and 8). The transformation efficiency of the explants transformed with pGmGRF5-GmGIF1 was 2.5%, and five transgenic plants had mutations in the GmFAD2 target sites (FIGS. 6F and 6G, Tables 7, and 4). One transgene-free editing event was also identified (Tables 7 and 8). Evidently the GmGRF5-GmGIF1 complex greatly stimulates regeneration and hence genetic transformation and genome editing in marginally transformable soybean cultivars. Also, the GmGRF5-GmGIF1 complex is able to generate transgene-free mutants due to transient expression of the CRISPR DNA and the fact that expression of the booster genes stimulated much more extensive cell proliferation, while the selection pressure was relatively low.

Example 10. The Soybean GmGRF5-GmGIF1 Complex Promotes Regeneration of Strawberry and Genome Editing (127) The regeneration of another dicot, strawberry (*Fragaria ananassa*), is genotype-dependent. Transformation of one of the most widely cultivated octoploid strawberry cultivars, Benihoppe, is extremely difficult, and no genome editing events have yet been obtained in this cultivar (Folta and Dhingra, 2006).

(128) The inventors examined the regeneration and genome editing of Benihoppe using the soybean GmGRF5-GmGIF1 complex because strawberry GRF proteins are close to soybean GRFs (FIG. 11A). The fusion protein was co-expressed with Cas9 in a single expression cassette, in vector pHUE411-GFP-GmGRF5-GmGIF1 (FIG. 7A). The pectate lyase gene, FaPL, which contributes to loss of strawberry fruit firmness (Jimenez-Bermudez et al., 2002), was targeted by Cas9 (FIG. 11B) and green fluorescent protein (GFP) was used as a visual reporter that allowed continuous monitoring of transgenic events. The control vector (pHUE411-GFP) contained Cas9, the sgRNA targeting FaPL, and the GFP gene. A total of 541 and 540 strawberry leaf explants were infected by the *Agrobacterium* contained pHUE411-GFP-GmGRF5-GmGIF1 and the control vector, respectively (FIG. 7B). Calli from leaf explants induced on callus induction medium were screened for GFP expression three weeks after infection. 19.6% (106/541) of the explants transformed with GmGRF5-GmGIF1 produced at least one GFP positive callus, compared with 6.5% (35/540) in the control (FIG. 7B).

(129) Moreover, the leaf explants transformed with GmGRF5-GmGIF1 tended to form multiple and large calli with strong GFP fluorescence, whereas, only a few small GFP-positive spots were observed on callus masses in the control (FIG. 7C). During growth on shoot induction medium the calli on explants transformed with GmGRF5-GmGIF1 grow more vigorously than those on the control (FIG. 7D), and shoot initiation occurred on 8.1% of the former calli from explants infected with GmGRF5-GmGIF1 five weeks after infection compared with only 3.0% of the latter (FIG. 7D). The transformation efficiency of the explants transformed with GmGRF5-GmGIF1 was 0.9%, and mutations in the FaPL target sites were detected in 0.6% of the explants (FIG. 7B, FIG. 11C), whereas neither transgenic plants and nor mutant plants were detected in the control (FIG. 7B). These data demonstrate that the GmGRF5-GmGIF1 complex could promote the regeneration, genetic transformation and genome editing of a transformation-recalcitrant strawberry cultivar, showing that GmGRF5-GmGIF1 stimulates genetic transformation and genome editing in other dicot plants. Example 11. Comparison of the Effects of Wild Type TaGRF4-TaGIF1 and Mutated TaGRF4-TaGIF1 on Regeneration and Genome Editing of Common Wheat

(130) To broaden the application of GRF-GIF1 complexes, the inventors examined their effect in the monocot, common wheat, in which transformation is limited to a narrow range of genotypes (He et al., 2015, Jones et al., 2005). Phylogenetic tree analysis suggested that the soybean GRF family, including GmGRF5, is distantly related to the monocot GRF family of rice (Chen et al., 2019). Previous work has also shown that overexpression of OsGRF4 and its coactivator OsGIF1 stimulates the cell proliferation and seed size in rice (Sun et al., 2016; Hu et al., 2015). Therefore, the inventors tested the effects of the rice homologs, TaGRF4, and TaGIF1, in common wheat (FIG. 12) while adding two modifications.

(131) First, TaGRF4 was fused with TaGRF1 to form a TaGRF4-TaGIF1 complex, and the miR396 target site in

TaGRF4 was inactivated by five point mutations (mTaGRF4-TaGIF1) to increase transcription of TaGRF4 (FIGS. 8A and 8B). Second, a transient expression strategy (Liang et al., 2017; Zhang et al., 2016; Zhang et al., 2018), which avoids any selection pressure throughout the whole tissue culture procedure, was used to generate transgene-free mutants in the TO generation (FIG. 8C). The wild type complex TaGRF4-TaGIF1, and mutant complex, mTaGRF4-TaGIF1, were separately transformed into the efficiently transformable common wheat cultivars Bobwhite and Kenong 199 (FIGS. 8B and 8C), together with the cytosine base editor pUBI-A3A vector containing an sgRNA for the common wheat acetolactate synthase gene (TaALS) (FIG. 13), in which C-to-T substitutions at Pro197 confers resistance to nicosulfuron (Zhang et al., 2019; Zong et al., 2018). The combination of pUBI-A3A and pUBI-GFP served as a control. After six weeks on non-selective media, numbers of regenerated plants and TaALS mutations were analyzed.

(132) The inventors found that immature embryos of both Bobwhite and Kenong 199 transformed with TaGRF4-TaGIF1 produced more regenerated plants (508.0% and 654.5%, respectively) than the control groups (81.0% and 136.9%, respectively) (FIG. 8D, Table 11). Moreover, mTaGRF4-TaGIF1 gave even higher regeneration ratios of 630.1% and 1165.4%, in Bobwhite and Kenong 199, respectively, 1.2- and 1.7-fold higher than the regeneration frequency with the wild type TaGRF4-TaGIF1 complex, and 7.8- and 8.5-fold higher than with the control construct (FIG. 8D, Tables 11 and 12). The inventors also examined genome editing at the TaALS target site using PCR-RE assays and Sanger sequencing. Consistent with the regeneration ratios, the Bobwhite and Kenong 199 plants transformed with the TaGRF4-TaGIF1 complex had higher mutation frequencies (32.7% and 103.4%, respectively) than plants transformed with the control construct (9.9% and 17.7%, respectively) (FIGS. 8E and 8D, Tables 11 and 12). Among the plants generated by the mTaGRF4-TaGIF1 complex the inventors identifies 216 mutants in 343 embryos (63.3%) and 577 mutants in 328 embryos (176.1%) in Bobwhite and Kenong 199, respectively, 1.7-1.9-fold higher than in the plants generated by the TaGRF4-TaGIF1 complex, and 6.4-9.9-fold higher than in those in the control groups (FIG. 8E, Tables 11 and 12).

(133) Since the plasmids were delivered using a transient expression approach (FIG. 8C), there seemed a high probability that the TaGRF4-TaGIF1 and base editor DNA constructs had not been integrated into the genome of the mutant plants. To test for the presence of plasmid DNAs in the regenerated TO mutants the inventors used a total of six primer sets (three for pUBI-A3A, three for pTaGRF4-TaGIF1 or pmTaGRF-TaGIF1) to amplify different regions of the TaGRF4-TaGIF1 and the base editor constructs, together covering almost the entire constructs (FIG. 8F). Based on the PCR analysis, the two vectors (pUBI-A3A and pTaGRF4-TaGIF1/pmTaGRF4-TaGIF1) were absent from 47.1%-55.4% of the mutants of Bobwhite and Kenong 199 treated with TaGRF4-TaGIF1, mTaGRF4-TaGIF1 and the control construct (FIG. 8G, Tables 11 and 12). The total numbers of transgene-free mutants in the mTaGRF4-TaGIF1-treated groups were 6.0-11.6-fold higher than in the control groups (Tables 11 and 12). Moreover, when the inventors grew the transgene-free mutants of Kenong 199 derived from mTaGRF4-TaGIF1 in a greenhouse the inventors found that they were fertile and displayed no obvious phenotypic differences from WT plants throughout all of development (FIG. 8H).

(134) In summary, the TaGRF4-TaGIF1 complex increased both regeneration frequency and genome-editing efficiency in common wheat and disruption of the miR396 target site further enhanced its efficiency. Furthermore, transient expression of mTaGRF4-TaGIF1 had no ill-effect on phenotype.

Example 12. The Mutated TaGRF4-TaGIF1 Complex Extends Genome Editing in Broad Common Wheat Cultivars

(135) The inventors tested whether the mTaGRF4-TaGIF1 complex improved regeneration rates and genome editing efficiencies in various common wheat cultivars that are widely grown in China, pUBI-A3A vector, containing an sgRNA for TaALS, was delivered with mTaGRF4-TaGIF1 into immature embryos of nine major Chinese common wheat cultivars Jimai 20, Jimai 22, Jing 411, Shannong 20, Shannong 116, Xiaoyan 54, Zhoumai 27, Zhoumai 28 and Zhongmai 175 according to the procedure in FIG. 8C. The combination of pUBI-A3A and pUBI-GFP served as a control.

(136) After six weeks on non-selective medium, the inventors found that mTaGRF4-TaGIF1 stimulated regeneration of these common wheat cultivars, and regeneration frequencies, which ranged from 9.9%-440.8%, were significantly higher than in the corresponding controls (0%-187.3%) (FIGS. 9A and 9B, Tables 13 and 14). In particular, mTaGRF4-TaGIF1 provoked substantial the regeneration and genome editing of Xiaoyan 54, Zhoumai 28, Jimai 20, Jimai 22 and Shannong 20, which is normally very difficult (FIGS. 9A and 9B, Table 13). Mutations in the TaALS sites were detected by PCR-RE assays and Sanger sequencing in all of these nine common wheat cultivars transformed with mTaGRF4-TaGIF1. Mutation frequencies ranged from 1.2-8.1%, compared with 0-3.9% in the control groups, five of which contained no mutants (FIG. 5C, Table 8). Moreover, homozygous mutant frequencies (HMF) in the nine cultivars transformed with mTaGRF4-TaGIF1 ranged from 10.0-55.6%, which was much higher than in the controls (0-15.4%) (Tables 13 and 15). The inventors also identified 22.2-66.7% transgene-free mutants among the nine cultivars transformed with the mTaGRF4-TaGIF1 complex (Table 13). In contrast, transgene-free mutants were only found in Shannong 116, Zhoumai 27 and Zhongmai 175 transformed with the control plasmid (Table 13). Thus, the mTaGRF4-TaGIF1 complex stimulates regeneration and genome editing in all nine common wheat cultivars.

(137) Soybean is one of the most important sources of edible oils and proteins, but its transformation rates remain very low despite the many available genetic transformation methods (Christou, 1992; Trick and Finer, 1998; Yang et al., 2016). Thus, the development of an improved transformation system for soybean is urgently needed. Plant GRF and GIF genes are highly expressed in meristematic tissues, including leaf and floral organ primordia and shoot apical meristems (Kim, 2019; Omidbakhshfard et al., 2015; Zhang et al., 2018), and might potentially be developed as boosters to stimulate plant regeneration so enhance genetic transformation and genome editing rates. The inventors screened and tested four GmGRF-GmGIF1 complexes in Williams 82, and found a transformation rate of 21.8% using the GmGRF5-GmGIF1 complex (FIG. 5I, Table 7), which is much higher than the transformation rate obtained using traditional transformation methods (Finer and McMullen, 1991; Hinchey et al., 1988). It should be noted that GmGRF5 is predicated as GmGRF7 in GenBank (XM_003526701.4), since it is most similar to *Arabidopsis* AtGRF7. (138) The genotype limitation depends on the susceptibility to *Agrobacterium* infection of the donor genotype and the regeneration ability of the explant (Jia et al., 2015). Fortunately, the GmGRF5-GmGIF1 complex improved the transformation efficiencies of Zhonghuang 13 (18.2%) and Hefeng 25 (2.5%), which are the most widely grown cultivars in China but are difficult to engineer genetically by current methods. This indicates that the GmGRF5-GmGIF1 complex can overcome the bottleneck of genotype-dependence in soybean transformation. GmGRF5-GmGIF1 enhanced recovery of transgene-free FAD2-edited events due to large number of regenerated shoots and use of a moderate stringency selection which promotes regenerated shoots as well as transgenic plant to elongate. This simulation of shoot regeneration by the GmGRF5-GmGIF1 complex and the resulting improved transformation and genome editing efficiencies in soybean and strawberry were further confirmed by the ability of soybean GROWTH-REGULATING FACTOR 5 (GmGRF5) and its cofactor soybean GIF1 complex to enhance regeneration, transformation and genome editing in diverse dicotyledonous plants.

(139) Considering their extremely wide distributions in all land plants, and their preferential expression in meristematic tissue (Kim, 2019; Omidbakhshfard et al., 2015; Shimano et al., 2018), native GRF-GIF1 complexes could be effective in promoting regeneration in most plants. The regeneration of monocotyledonous common wheat remains genotype-dependent (He et al., 2015; Jones et al., 2005), but the inventors demonstrated that transient expression of wheat mTaGRF4-TaGIF1 dramatically improved regeneration and genome editing, and also greatly increased numbers of transgene-free mutants in nine recalcitrant Chinese commercial elite cultivars, most of which had never been regenerated and gene-edited before. The improved TaGRF4-TaGIF1 complex (mTaGRF4-TaGIF1) thus overcomes the genotype limitations to regeneration in common wheat and boosts its genome editing.

(140) An ideal booster should not influence the morphology of regenerated plants. In the present work, no morphological changes were observed in the transgenic soybean plants, even when that the GmGRF5-GmGIF1 complex was constitutively expressed, probably due to the moderate level of its expression, which is regulated at the post-transcriptional level (Kim and Tsukaya, 2015; Liu et al., 2009; Rodriguez et al., 2010). In common wheat, the inventors found that the mTaGRF4-TaGIF1 complex performed better in improving the regeneration ability and genome editing frequency than the original TaGRF4-TaGIF1 complex. Although mTaGRF4-TaGIF1 was not repressed by miR396, the transient expression system prevented prolonged presence of this complex in the plant cells, this not only minimized morphological side effects but also generated a large number of transgene-free mutants.

(141) In summary, the GmGRF5-GmGIF1 complex enhances the regeneration of dicotyledonous soybean and strawberry and hence stimulates genetic transformation and genome editing. Similarly, transient expression of the improved TaGRF4-TaGIF1 complex containing the inactivated miR396 target site stimulates regeneration and genome editing in monocotyledonous common wheat. Moreover, the GRF-GIF1 complexes are genotype-independent, as they work well in diverse soybean and common wheat cultivars. Given that members of the GRF gene family exists in many plant species, the GRF-GIF1 complexes described here holds great promise for improving genome-editing efficiency in a wide range of crop plants.

(142) TABLE-US-00007 TABLE 7 Effect of GmGRF-GmGIF1 fusions on soybean transformation and genome editing efficiency. No. elongated No. elongated putative putative taglufosinate- No. No. explants glufosinate- resistant Targeted Soybean No. explants with multiple resistant shoots shoots (≥ 9 cm gene cultivar Treatment explants recovered buds/RF (%) (≥ 2 cm in length) in length) GmFAD2 Williams 82 pBSE401 (CK) 450 412 263 (58.4 ± 5.8) 468 (156.0 ± 14.5) 101 (33.7 ± 4.9) (Knockout) GmGRF5-GmGIF1 450 432 422 (93.8 ± 4.0).sup.b 1297 (432.3 ± 18.5) 256 (85.3 ± 7.5) GmGRF6-GmGIF1 450 424 290 (64.4 ± 3.8) 413 (137.7 ± 14.5) 107 (35.7 ± 6.5) GmGRF11-GmGIF1 450 422 269 (59.8 ± 3.0) 463 (154.3 ± 8.1) 108 (36.0 ± 7.6) GmGRF18-GmGIF1 450 419 379 (67.6 ± 7.2) 520 (173.3 ± 5.8) 124 (41.3 ± 7.3) Zhonghuang 13 pBSE401 (CK) 450 435 295 (65.6 ± 5.1) 699 (233.0 ± 15.1) 97 (32.3 ± 3.1) GmGRF5-GmGIF1 450 440 378 (84.0 ± 2.7).sup.b 938 (312.6 ± 25.5) 205 (68.3 ± 5.5) Hefeng 25 pBSE401 (CK) 450 372 243 (54.0 ± 2.9) 0 0 GmGRF5-GmGIF1 450 372 320 (71.1 ± 8.8).sup.a 164 (54.7 ± 11.2) 36 (12.0 ± 3) Growth rate Average elongated No. transgenic No. No. transgene-free Targeted (days) shoots per explant plants/TE (%) mutants/MF (%) mutants/TEF (%) gene 70.0 $\pm$ 4.0 1.1 $\pm$ 0.1 38 (8.5 ± 2.0) 32 (7.1 ± 2.0) 0 GmFAD2 57.6 $\pm$ 2.5.sup.a 3.0 $\pm$ 0.2.sup.c 98 (21.8 ± 2.0).sup.b 94 (20.8 ± 3.7) 15 (5.0 ± 2.0) (Knockout) 70.3 $\pm$ 2.1 1.0 $\pm$ 0.1 37 (8.2 ± 1.6) 32 (7.1 ± 0.8) 0 70.0 $\pm$ 1.0 1.1 $\pm$ 0.1 38 (8.4 ± 3.7) 33 (7.3 ± 3.4) 0 69.3 $\pm$ 0.6 1.2 $\pm$ 0.1 46 (10.2 ± 1.0) 36 (8.0 ± 0.7) 0 89.3 $\pm$ 0.6 1.7 $\pm$ 0.1 24 (5.3 ± 0.7) 11 (2.4 ± 1.4) 0 87.3 $\pm$ 0.66 2.6 $\pm$ 0.2.sup.b 82 (18.2 ± 2.0).sup.b 72

(16.0 ± 1.3) 9 1 1.0 0 1.0 0 73.7 ± 3.7. sup.c 1 1 (2.3 ± 0.4). sup.b 6 1 (0.2 ± 0.4) RF (regeneration frequency) = no. of explants with multiple buds/total explant number × 100%. Growth rates were measured from the date of the initial transformation to the date when the first elongated shoot reached 9 cm in length. TE (transformation efficiency) = no. of transgenic plants/total explants × 100%. Elongated shoots (≥9 cm in length) were collected for transgenic assay only once on day 75 after *Agrobacterium*-mediated transformation. MF (mutation frequency) = no. of mutants/total transgenic plants × 100%. TFF (transgene-free frequency) = no. of transgene-free mutants/total mutants × 100%. The numbers and means for each treatment were calculated from data collected from three replicates of the experiment. Data are mean ± SD (n = 3). .sup.a,b,c indicate significant differences compared with the CK group (two-sided Student's t-test. .sup.a indicates P < 0.05; .sup.b indicates P < 0.01; .sup.c indicates P < 0.001).

(143) TABLE-US-00008 TABLE 8 Effect of the GmGRF-GmGIF1 fusions on soybean transformation and genome editing efficiency (Williams 82, Hefeng 25 and Zhonghuang 13, raw data). No. elongated No. putative putative explants glufosinate- glufosinate- with resistant resistant No. multiple shoots shoots Targeted Soybean No. explants buds/RF (≥2 cm (≥9 cm gene cultivar Repeat Treatment explants recovered (%) in length) in length) GmFAD2 Williams 82 I pBSE401 (CK) 150 138 97 (64.7) 155 36 (Knockout) GmGRF5-GmGIF1 150 147 147 (98.0) 432 85 GmGRF6-GmGIF1 150 145 92 (61.3) 138 36 GmGRF11-GmGIF1 150 141 85 (56.7) 145 35 GmGRF18-GmGIF1 150 136 106 (70.7) 180 44 II pBSE401 (CK) 150 140 86 (57.3) 171 37 GmGRF5-GmGIF1 150 142 135 (90.0) 451 93 GmGRF6-GmGIF1 150 144 103 (68.7) 123 42 GmGRF11-GmGIF1 150 143 90 (60.0) 159 44 GmGRF18-GmGIF1 150 141 89 (59.3) 170 47 III pBSE401 (CK) 150 134 80 (53.3) 142 28 GmGRF5-GmGIF1 150 143 140 (93.3) 414 78 GmGRF6-GmGIF1 150 135 95 (63.3) 152 29 GmGRF11-GmGIF1 150 138 94 (62.7) 159 29 GmGRF18-GmGIF1 150 142 109 (72.7) 170 33 Zhonghuang 13 I pBSE401 (CK) 150 145 90 (60.0) 216 35 GmGRF5-GmGIF1 150 147 126 (84.0) 300 68 II pBSE401 (CK) 150 141 105 (70.0) 245 29 GmGRF5-GmGIF1 150 149 122 (81.3) 342 63 III pBSE401 (CK) 150 149 100 (66.7) 238 33 GmGRF5-GmGIF1 150 144 130 (86.7) 296 74 Hefeng 25 I pBSE401 (CK) 150 121 83 (55.3) 0 0 GmGRF5-GmGIF1 150 132 121 (80.7) 45 9 II pBSE401 (CK) 150 128 84 (56.0) 0 0 GmGRF5-GmGIF1 150 123 95 (63.3) 52 15 III pBSE401 (CK) 123 76 (50.7) 0 0 GmGRF5-GmGIF1 117 104 (69.3) 67 12 Average No. No. Targeted Growth elongated shoots No. transgenic mutants/MF transgene-free gene rate (days) per explant plants/TE (%) (%) mutant/TEF (%) GmFAD2 70 1.1 12 (8.0) 10 (6.7) 0 (Knockout) 55 2.9 30 (20.0) 26 (17.3) 3 68 1.0 11 (7.3) 10 (6.7) 0 69 1.0 13 (8.7) 11 (7.3) 0 69 1.3 17 (11.3) 13 (8.7) 0 66 1.2 16 (10.7) 14 (9.3) 0 58 3.2 36 (24.0) 37 (24.6) 7 71 0.9 15 (10.0) 12 (8.0) 0 71 1.1 18 (12.0) 16 (10.7) 0 69 1.2 14 (9.3) 12 (8.0) 0 74 1.1 10 (6.7) 8 (5.3) 0 60 2.9 32 (21.3) 31 (20.6) 5 72 1.1 11 (7.3) 10 (6.7) 0 70 1.2 7 (4.6) 6 (4.0) 0 70 1.2 15 (10) 11 (7.3) 0 89 1.7 7 (4.7) 3 (2.0) 0 87 2.5 24 (16.0) 22 (14.7) 2 89 1.8 9 (6.0) 2 (1.3) 0 87 2.7 28 (18.7) 24 (16.0) 4 90 1.8 8 (5.3) 6 (4.0) 0 88 2.5 30 (20.0) 26 (17.3) 3 1 0.0 0 0 0 70 0.4 4 (2.7) 1 (0.7) 0 — 0.0 0 0 0 76 0.5 4 (2.7) 2 (1.3) 1 1 0.0 0 0 0 75 0.5 3 (2.0) 3 (2.0) 0 RF (regeneration frequency) = no. of explants with multiple buds/total explant number × 100%. Growth rates were measured from the date of the initial transformation to the date when the first elongated shoot reached 9 cm in length. TE (transformation efficiency) = no. of transgenic plants/total explants × 100%. Elongated shoots (≥9 cm in length) were collected for transgenic assay only once on day 75 after *Agrobacterium*-mediated transformation. MF (mutation frequency) = no. of mutants/total transgenic plants × 100%. TFF (transgene-free frequency) = no. of transgene-free mutants/total mutants × 100%.

(144) TABLE-US-00009 TABLE 9 Mutation identification using PCR-RE assays for mutants regenerated from Williams 82 explants transformed with pGmGRF5-GmGIF1, pGmGRF6-GmGIF1, pGmGRF11-GmGIF1, pGmGRF18-GmGIF1, and pBSE401, respectively. Vector Mutant ID Genotype Mutants ID Genotype Mutants ID Genotype pGmGRF5-GmGIF1 Gm-001 AaBb Gm-033 AaBb Gm-065 AaBB Gm-002 AaBb Gm-034 AaBB Gm-066 AaBb Gm-003 AaBb Gm-035 AaBb Gm-067 AaBb Gm-004 AaBb Gm-036 AaBB Gm-068 AaBB Gm-005 AaBb Gm-037 aabb Gm-069 AaBb Gm-006 AaBb Gm-038 AaBb Gm-070 aabb Gm-007 AaBB Gm-039 AaBb Gm-071 AaBb Gm-008 AaBb Gm-040 AaBb Gm-072 AaBb Gm-009 AaBb Gm-041 AaBb Gm-073 AaBb Gm-010 AaBB Gm-042 AaBb Gm-074 AaBb Gm-011 AaBb Gm-043 AaBb Gm-075 AaBb Gm-012 aabb Gm-044 AaBb Gm-076 AaBb Gm-013 AaBb Gm-045 AaBB Gm-077 AaBb Gm-014 AaBb Gm-046 AaBb Gm-078 AaBB Gm-015 AaBb Gm-047 AaBB Gm-079 AaBb Gm-016 AaBb Gm-048 AaBb Gm-080 AaBB Gm-017 AaBb Gm-049 AaBb Gm-081 aabb Gm-018 AaBb Gm-050 AaBb Gm-082 AaBB Gm-019 AaBb Gm-051 AaBb Gm-083 AaBb Gm-020 AaBB Gm-052 AaBb Gm-084 AaBb Gm-021 AaBb Gm-053 AaBb Gm-085 AaBb Gm-022 AaBB Gm-054 AaBB Gm-086 AaBb Gm-023 aabb Gm-055 AaBb Gm-087 AaBb Gm-024 AaBb Gm-056 AaBB Gm-088 AaBb Gm-025 AaBb Gm-057 aabb Gm-089 AaBB Gm-026 AaBb Gm-058 AaBb Gm-090 AaBb Gm-027 AaBb Gm-059 AaBb Gm-091 AaBb Gm-028 AaBb Gm-060 AaBb Gm-092 AaBB Gm-029 AaBB Gm-061 AaBb Gm-093 AaBb Gm-030 AaBb Gm-062 AaBb Gm-094 AaBb Gm-031 AaBB Gm-063 AaBb Gm-032 aabb Gm-064 AaBB Gm-101 AaBb Gm-112 AaBb Gm-123 AaBB Gm-102 AaBB Gm-113 AaBb Gm-124 aabb Gm-103 AaBB Gm-114 AaBB Gm-125 AaBb Gm-104 AaBb Gm-115 AaBb Gm-126 AaBb pGmGRF6-GmGIF1 Gm-105 AaBb Gm-116 AaBb Gm-127 AaBb Gm-106 AaBB Gm-117 AaBB Gm-128 AaBb Gm-107 AaBB Gm-118 AaBB Gm-129 AaBb Gm-108 AaBb Gm-119 AaBb Gm-130 AaBb Gm-109 AaBb Gm-120 AaBb Gm-131 AaBb Gm-110 AaBb Gm-121 AaBb Gm-132 AaBb Gm-111 AaBB Gm-122 AaBb pGmGRF11-GmGIF1 Gm-201 AaBb Gm-212 AaBb Gm-223 AaBB Gm-202 AaBB Gm-213 AaBB Gm-224 AaBb Gm-203 AaBb Gm-214 AaBb Gm-225 AaBB Gm-204 AaBB Gm-215 AaBB Gm-226 AaBb Gm-205 AaBB Gm-216

Gm-227 AaBb Gm-206 AaBb Gm-217 AaBb Gm-228 AaBb Gm-207 AaBb Gm-218 AaBb Gm-229 AaBb Gm-208 AaBb Gm-219 AaBb Gm-230 AaBb Gm-209 AaBb Gm-220 AaBb Gm-231 AaBb Gm-210 AaBb Gm-221 AaBb Gm-232 AaBb Gm-211 AaBB Gm-222 AaBb Gm-233 AaBb pGmGRF18-GmGIF1 Gm-301 AaBb Gm-313 AaBb Gm-325 AaBb Gm-302 AaBb Gm-314 AaBb Gm-326 AaBB Gm-303 AaBB Gm-315 AaBb Gm-327 AaBb Gm-304 AaBb Gm-316 AaBb Gm-328 AaBb Gm-305 AaBB Gm-317 AaBB Gm-329 AaBb Gm-306 AaBB Gm-318 AaBb Gm-330 AaBb Gm-307 AaBb Gm-319 AaBB Gm-331 AaBb Gm-308 AaBb Gm-320 AaBB Gm-332 AaBB Gm-309 AaBb Gm-321 AaBB Gm-333 AaBb Gm-310 AaBb Gm-322 AaBb Gm-334 AaBB Gm-311 AaBb Gm-323 AaBB Gm-335 AaBb Gm-312 AaBb Gm-324 AaBB Gm-336 AaBb pBSE401 (CK) Gm-401 AaBb Gm-413 AaBB Gm-425 AaBb Gm-402 AaBb Gm-414 AaBb Gm-426 AaBb Gm-403 AaBb Gm-415 AaBB Gm-427 AaBb Gm-404 AaBb Gm-416 AaBB Gm-428 AaBb Gm-405 AaBb Gm-417 AaBb Gm-429 AaBb Gm-406 AaBb Gm-418 AaBb Gm-430 AaBB Gm-407 AaBb Gm-419 AaBb Gm-431 AaBb Gm-408 AaBb Gm-420 AaBb Gm-432 AaBB Gm-409 AaBb Gm-421 AaBb Gm-410 AaBb Gm-422 AaBb Gm-411 AaBb Gm-423 AaBB Gm-412 AaBb Gm-424 AaBb Aa/aa and Bb/bb represent heterozygous/homozygous mutations in FAD2-1A and FAD2-1B, respectively.

(145) TABLE-US-00010 TABLE 10 Mutation identification using PCR-RE assays for mutants regenerated from explants (Zhonghuang 13 and Hefeng 25) transformed with pGmGRF5-GmGIF and pBSE401. Soybean cultivar Vector Mutant ID Genotype Mutant ID Genotype Mutant ID Genotype Hefeng 25 pGmGRF5- Gm-501 AaBb Gm-503 AaBB Gm-505 AaBB GmGIF Gm-502 AaBB Gm-504 AaBb Gm-506 AaBb Zhonghuang 13 pGmGRF5- Gm-601 AaBB Gm-625 AaBb Gm-649 AaBb GmGIF1 Gm-602 AaBb Gm-626 AaBb Gm-650 AaBB Gm-603 AaBb Gm-627 AaBb Gm-651 AaBb Gm-604 AaBb Gm-628 AaBB Gm-652 AaBB Gm-605 AaBB Gm-629 AaBb Gm-653 AaBB Gm-606 AaBb Gm-630 AaBB Gm-654 AaBb Gm-607 AaBB Gm-631 AaBB Gm-655 AaBb Gm-608 AaBB Gm-632 AaBb Gm-656 AaBb Gm-609 AaBb Gm-633 AaBb Gm-657 AaBb Gm-610 AaBb Gm-634 AaBb Gm-658 AaBb Gm-611 AaBb Gm-635 AaBb Gm-659 AaBB Gm-612 AaBb Gm-636 AaBb Gm-660 AaBb Gm-613 AaBb Gm-637 AaBB Gm-661 AaBB Gm-614 AaBB Gm-638 AaBb Gm-662 AaBB Gm-615 AaBb Gm-639 AaBB Gm-663 AaBb Gm-616 AaBB Gm-640 AaBB Gm-664 AaBb Gm-617 AaBB Gm-641 AaBb Gm-665 AaBb Gm-618 AaBb Gm-642 AaBb Gm-666 AaBb Gm-619 AaBb Gm-643 AaBb Gm-667 AaBb Gm-620 AaBb Gm-644 AaBb Gm-668 AaBB Gm-621 AaBb Gm-645 AaBb Gm-669 AaBb Gm-622 AaBb Gm-646 AaBB Gm-670 AaBB Gm-623 AaBB Gm-647 AaBb Gm-671 AaBB pBSE401 Gm-624 AaBb Gm-648 AaBB Gm-672 AaBb Gm701 AaBb Gm705 AaBb Gm709 AaBb Gm702 AaBb Gm706 AaBb Gm710 AaBB Gm703 AaBB Gm707 AaBb Gm711 AaBb Gm704 AaBb Gm708 AaBb Aa/aa and Bb/bb represent heterozygous/homozygous mutations in FAD2-1A and FAD2-1B, respectively.

(146) TABLE-US-00011 TABLE 11 Effect of the TaGRF4-TaGIF1 and mTaGRF4-TaGIF1 complexes on regeneration and genome editing efficiencies of common wheat cultivars Bobwhite and Kenong 199. Common No. transgene- Targeted wheat immature No. regenerated No. mutants/ free mutants/ gene cultivar Treatment embryos plants/ RF (%) MF (%) TFF (%) TaALS Bobwhite pUBI-GFP (CK) 333 266 (81.0 ± 19.8) 33 (9.9 ± 1.3) 17 (50.9 ± 5.5) (C to T) TaGRF4-TaGIF1 277 1507 (508.0 ± 75.9) 100 (32.7 ± 3.45) 47 (47.1 ± 5.9) mTaGRF4-TaGIF1 343 2080 (630.1 ± 86.2) 216 (63.3 ± 4.0) 103 (47.5 ± 2.6) pUBI-GFP (CK) 307 419 (136.9 ± 5.7) 55 (17.7 ± 2.6) 28 (52.8 ± 8.4) Kenong 199 TaGRF4-TaGIF1 325 2095 (654.5 ± 55.5) 331 (103.4 ± 9.3) 170 (51.6 ± 1.8) mTaGRF4-TaGIF1 328 3796 (1165.4 ± 58.8) 577 (176.1 ± 13.9) 324 (55.4 ± 6.6) RF (regeneration frequency) = no. of regenerated plants/total immature embryos × 100%. MF (mutation frequency) = no. of mutants/total immature embryos × 100%. The numbers and means for each treatment were calculated from data collected from three replicates of the experiment. Data are means ± s.e.m (n = 3).

(147) TABLE-US-00012 TABLE 12 Effect of the TaGRF4-TaGIF1 and mTaGRF4-TaGIF1 complexes on regeneration and genome editing efficiencies of common wheat cultivars Bobwhite and Kenong 199 (raw data). No. No. Common transgene- Targeted wheat immature regenerated mutants/MF free mutants/ gene cultivar Treatment Repeat embryos plants/RF (%) (%) TFF (%) TaALS Bobwhite pUBI-GFP (CK) 1 119 50 (42.0) 10 (8.4) 4 (40.0) (C to T) 2 112 119 (106.2) 14 (12.5) 8 (57.1) 3 102 97 (95.0) 9 (8.8) 5 (55.6) TaGRF4-TaGIF1 1 116 547 (472.6) 42 (25.9) 20 (47.6) 2 86 562 (653.5) 30 (34.9) 11 (36.7) 3 75 398 (530.7) 28 (37.3) 16 (57.1) mTaGRF4-TaGIF1 1 103 498 (483.5) 70 (68.0) 32 (45.7) 2 117 915 (782.1) 78 (66.7) 41 (52.6) 3 123 767 (624.6) 68 (55.3) 30 (44.2) Kenong pUBI-GFP (CK) 1 102 146 (143.1) 17 (16.7) 7 (41.2) 199 2 110 138 (125.5) 25 (22.7) 12 (48.0) 3 95 135 (142.1) 13 (13.7) 9 (69.2) TaGRF4-TaGIF1 1 132 763 (578.0) 116 (87.9) 56 (48.3) 2 100 623 (623.0) 120 (120) 62 (51.7) 3 93 709 (762.4) 95 (102.2) 52 (54.7) mTaGRF4-TaGIF1 1 114 1329 (1165.8) 230 (201.8) 135 (58.7) 2 120 1276 (1063.3) 185 (154.2) 120 (64.9) 3 94 1191 (1267.0) 162 (172.3) 69 (42.6) RF (regeneration frequency) = no. of regenerated plants/total immature embryos × 100%. MF (mutation frequency) = no. of mutants/total immature embryos × 100%.

(148) TABLE-US-00013 TABLE 13 Effect of mTaGRF4-TaGIF1 on regeneration and genome editing efficiencies of nine elite common wheat cultivars. Common No. No. No. transgene- Targeted wheat bombarded No. regenerated mutants/ No. homozygous free mutants/ gene cultivar Treatment embryos plants/ RF (%) MF (%) mutants/HMF (%) TFF (%) TaALS Jimai 20 UBI-GFP (CK) 335 14 (4.2 ± 1.1) 0 0 0 (C-to-T) Jimai 22 mTaGRF4-TaGIF1 270 37 (14.2 ± 2.1) 9 (3.3) 5 (55.6) 2 (22.2) Jing 411 UBI-GFP (CK) 272 13 (4.7 ± 0.6) 0 0 0 Shannong 20 mTaGRF4-TaGIF1 344 34 (9.9 ± 0.4) 4 (1.2) 0 1 (25.0) Shannong 116 UBI-GFP (CK) 203 65 (18.8 ± 2.7) 2

(1.0) 0 0 Xiaoyan 54 mTaGRF4-TaGIF1 320 135 (34.0 ± 3.2) 5 (1.6) 1 (25.0) Zhoumai 27 UBI-GFP (CK) 335 15 (4.5 ± 0.6) 0 0 0 Zhoumai 28 pmTaGRF4-TaGIF1 344 71 (20.7 ± 2.2) 10 (2.9) 1 (10.0) 5 (50.0) Zhongmai 175 UBI-GFP (CK) 285 128 (66.3 ± 8.2) 3 (1.1) 0 1 (33.3) mTaGRF4-TaGIF1 308 342 (101.7 ± 18.4) 7 (2.3) 1 (14.3) 2 (28.6) UBI-GFP (CK) 320 0 0 0 0 mTaGRF4-TaGIF1 343 36 (11.0 ± 2.1) 6 (1.7) 1 (16.7) 3 (50.0) UBI-GFP (CK) 286 441 (157.6 ± 24.1) 4 (1.4) 0 (0.0) 1 (25.0) mTaGRF4-TaGIF1 343 1578 (440.8 ± 50.3) 8 (2.3) 1 (12.5) 5 (62.5) UBI-GFP (CK) 308 16 (5.2 ± 1.3) 0 0 0 mTaGRF4-TaGIF1 331 370 (112.3 ± 11.4) 10 (3.0) 2 (20.0) 4 (40.0) UBI-GFP (CK) 335 614 (187. ± 26.1) 13 (3.9) 2 (15.4) 8 (61.6) mTaGRF4-TaGIF1 332 1265 (380.6 ± 29.0) 27 (8.1) 10 (37.0) 18 (66.7) RF (regeneration frequency) = no. of regenerated plants/total embryos × 100%. MF (mutation frequency) = no. of mutants/total immature embryos × 100%. HMF (homozygous mutant frequency) = no. of homozygous mutants/total mutants × 100%. TFF (transgene-free frequency) = no. of transgene-free mutants/total mutants × 100%. The numbers and means for each treatment were calculated from data collected from three replicates of the experiment. Data are means ± s.e.m (n = 3).

(149) TABLE-US-00014 TABLE 14 Effect of expression of the mutated common wheat TaGRF4-TaGIF1 on the regeneration and genome editing efficiencies of nine elite common wheat cultivars (raw data). Common wheat No. bombarded No. regenerated cultivar Treatment Repeat immature embryos plants/RF (%) Jimai 20 pUBI-GFP (CK) 1 115 4 (3.5) 2 110 7 (6.4) 3 110 3 (2.7) mTaGRF4-TaGIF1 1 109 11 (10.1) 2 91 14 (15.4) 3 70 12 (17.1) Jimai 22 pUBI-GFP (CK) 1 102 6 (5.9) 2 66 3 (4.5) 3 104 4 (3.8) mTaGRF4-TaGIF1 1 118 11 (9.3) 2 123 13 (10.6) 3 103 10 (9.7) Jing 411 pUBI-GFP (CK) 1 115 20 (17.4) 2 116 28 (24.1) 3 113 17 (15.0) mTaGRF4-TaGIF1 1 130 40 (30.8) 2 140 43 (30.7) 3 128 52 (40.6) Shangnong 20 pUBI-GFP (CK) 1 114 5 (4.4) 2 115 4 (3.5) 3 106 6 (5.7) mTaGRF4-TaGIF1 1 115 19 (16.5) 2 116 25 (21.6) 3 113 27 (23.9) Shangnong 116 pUBI-GFP (CK) 1 60 49 (81.7) 2 75 40 (53.3) 3 61 39 (63.9) mTaGRF4-TaGIF1 1 133 124 (93.2) 2 90 110 (122.2) 3 85 108 (127.1) Xiaoyan 54 pUBI-GFP (CK) 1 98 0 2 108 0 3 114 0 mTaGRF4-TaGIF1 1 98 10 (10.2) 2 100 15 (15.0) 3 141 11 (7.8) Zhoumai 27 pUBI-GFP (CK) 1 81 160 (197.5) 2 105 120 (114.3) 3 100 161 (161.0) mTaGRF4-TaGIF1 1 109 500 (458.7) 2 132 457 (346.2) 3 120 621 (517.5) Zhoumai 28 pUBI-GFP (CK) 1 115 8 (7.0) 2 86 5 (5.9) 3 107 3 (2.8) mTaGRF4-TaGIF1 1 129 130 (100.8) 2 105 142 (135.2) 3 97 98 (101.0) Zhongmai 175 pUBI-GFP (CK) 1 95 198 (208.2) 2 110 240 (218.2) 3 130 176 (135.4) mTaGRF4-TaGIF1 1 120 389 (324.2) 2 107 450 (420.6) 3 105 417 (397.1) RF (regeneration frequency) = no. of regenerated plants/total immature embryos × 100%.

(150) TABLE-US-00015 TABLE 15 Genotypes of the T0 taals mutants regenerated from immature embryos of nine elite common wheat cultivars treated with mTaGRF4-TaGIF1 and UBI-GFP (CK), respectively. Wheat cultivar Vectors delivered Mutant ID Genotype A subgenome B subgenome D subgenome Transgene-free Zhongmai 175 pmTaGRF4- Ta-001 AaBbDd C.sub.6C.sub.7C.sub.8C.sub.9C.sub.10 > T.sub.6T.sub.7T.sub.8T.sub.9T.sub.10 C.sub.6C.sub.7C.sub.8C.sub.9C.sub.10 > T.sub.6T.sub.7T.sub.8T.sub.9T.sub.10 C.sub.6C.sub.7C.sub.8C.sub.9C.sub.10 > T.sub.6T.sub.7T.sub.8T.sub.9T.sub.10 YES TaGIF1 + pUBI-A3A C.sub.6C.sub.7C.sub.8C.sub.9 > T.sub.6T.sub.7T.sub.8T.sub.9 C.sub.6C.sub.7 > T.sub.6T.sub.7 Ta-002 aabbdd C.sub.6C.sub.7C8C9C.sub.10 > T.sub.6T.sub.7T.sub.8T.sub.9T.sub.10 C.sub.6C.sub.7C8C.sub.9C.sub.10 > T.sub.6T.sub.7T.sub.8T.sub.9T.sub.10 C.sub.6C.sub.7C.sub.8 > T.sub.6T.sub.7T.sub.8 YES Ta-003 AaBbdd C.sub.6C.sub.7C8C9C.sub.10 > T.sub.6T.sub.7T.sub.8T.sub.9T.sub.10 C.sub.6C.sub.7C.sub.8 > T.sub.6T.sub.7T.sub.8 C.sub.6C.sub.7C.sub.8C.sub.9C.sub.10 > T.sub.6T.sub.7T.sub.8T.sub.9T.sub.10 NO Ta-004 aabbdd C.sub.6C.sub.7C.sub.8C.sub.9C.sub.10 > T.sub.6T.sub.7T.sub.8T.sub.9T.sub.10 C.sub.6C.sub.7 > T.sub.6T.sub.7; Del CCGCC; YES C.sub.6C.sub.7C.sub.8C.sub.9 > T.sub.6T.sub.7T.sub.8T.sub.9 Del CGCC Ta-005 aabbdd C.sub.6C.sub.7C.sub.8 > T.sub.6T.sub.7T.sub.8 C.sub.6C.sub.7C.sub.8C.sub.9C.sub.10 > T.sub.6T.sub.7T.sub.8T.sub.9T.sub.10 C.sub.6C.sub.7C.sub.8 > T.sub.6T.sub.7T.sub.8; YES C.sub.6C.sub.7C.sub.8C.sub.9C.sub.10 > T.sub.6T.sub.7T.sub.8T.sub.9T.sub.10 Ta-006 AaBbDD C.sub.6C.sub.7C.sub.8C.sub.9C.sub.10 > T.sub.6T.sub.7T.sub.8T.sub.9T.sub.10 C.sub.6C.sub.7C.sub.8C.sub.9C.sub.10 > T.sub.6T.sub.7T.sub.8T.sub.9T.sub.10 WT YES Ta-007 AaBbDd C.sub.6C.sub.7C.sub.8C.sub.9C.sub.10 > T.sub.6T.sub.7T.sub.8T.sub.9T.sub.10 C.sub.6C.sub.7C.sub.8C.sub.9C.sub.10 > T.sub.6T.sub.7T.sub.8T.sub.9T.sub.10 C.sub.6C.sub.7 > T.sub.6T.sub.7 YES Ta-008 AaBBDD C.sub.6C.sub.7C.sub.8C.sub.9C.sub.10 > T.sub.6T.sub.7T.sub.8T.sub.9T.sub.10 WT WT NO Ta-009 aabbdd C.sub.6C.sub.7C.sub.8C.sub.9 > T.sub.6T.sub.7T.sub.8T.sub.9 C.sub.6C.sub.7C.sub.8 > T.sub.6T.sub.7T.sub.8 YES C.sub.6C.sub.7C.sub.8C.sub.9C.sub.10 > T.sub.6T.sub.7T.sub.8T.sub.9T.sub.10 C.sub.6C.sub.7C.sub.8C.sub.9C.sub.10 > T.sub.6T.sub.7T.sub.8T.sub.9T.sub.10 Ta-010 aabbDd C.sub.6C.sub.7C.sub.8 > T.sub.6T.sub.7T.sub.8 C.sub.6C.sub.7C.sub.8C.sub.9C.sub.10 > T.sub.6T.sub.7T.sub.8T.sub.9T.sub.10 C.sub.6C.sub.7 > T.sub.6T.sub.7 YES Ta-011 AaBbDD AGGTCCCCCGCCGC > TCACGGGCCAGGTCCCC > NO GCCCC ATCCTGTTTCGGGCCAGGT Ta-012 AaBbDD Del AGGTCCCCC; WT CCCCC > TGTTT-----; WT YES Ta-013 aabbdd C.sub.6C.sub.7C.sub.8C.sub.9C.sub.10 > T.sub.6T.sub.7T.sub.8T.sub.9T.sub.10 C.sub.6C.sub.7C.sub.8C.sub.9C.sub.10 > T.sub.6T.sub.7T.sub.8T.sub.9T.sub.10 NO C.sub.6C.sub.7C.sub.8C.sub.9 > T.sub.6T.sub.7T.sub.8T.sub.9 C.sub.6C.sub.7C.sub.8 > T.sub.6T.sub.7T.sub.8 Ta-014 AaBbdd C.sub.6C.sub.7C.sub.8C.sub.9C.sub.10 > T.sub.6T.sub.7T.sub.8T.sub.9T.sub.10

[illegible]

[illegible]

C.sub.6C.sub.8C.sub.10 > T.sub.6T.sub.7T.sub.8T.sub.9T.sub.10 C.sub.6C.sub.7C.sub.8C.sub.9 > T.sub.6T.sub.7T.sub.8T.sub.9; NO C.sub.6 > T.sub.6 Jimai 22 pmTaGRF4- Ta-171 AaBbDd
C.sub.6C.sub.7C.sub.8C.sub.9C.sub.10 > T.sub.6T.sub.7T.sub.8T.sub.9T.sub.10
C.sub.6C.sub.7C.sub.8C.sub.9C.sub.10 > T.sub.6T.sub.7T.sub.8T.sub.9T.sub.10
C.sub.6C.sub.7C.sub.8C.sub.9C.sub.10 > T.sub.6T.sub.7T.sub.8T.sub.9T.sub.10 NO TaGIF1 + pUBI-A3A Ta-172 AaBbDd C.sub.6C.sub.7C.sub.8C.sub.9C.sub.10 > T.sub.6T.sub.7T.sub.8T.sub.9T.sub.10
C.sub.6C.sub.7C.sub.8C.sub.9C.sub.10 > T.sub.6T.sub.7T.sub.8T.sub.9T.sub.10
C.sub.6C.sub.7C.sub.8C.sub.9C.sub.10 > T.sub.6T.sub.7T.sub.8T.sub.9T.sub.10; YES Ta-173 AaBBDd
C.sub.6C.sub.7C.sub.8C.sub.9C.sub.10 > T.sub.6T.sub.7T.sub.8T.sub.9T.sub.10 WT
C.sub.6C.sub.7C.sub.8C.sub.9C.sub.10 > T.sub.6T.sub.7T.sub.8T.sub.9T.sub.10 NO Ta-174 AaBbDD
C.sub.6C.sub.7C.sub.8 > T.sub.6T.sub.7T.sub.8 C.sub.6C.sub.7C.sub.8 > T.sub.6T.sub.7T.sub.8 WT NO pUBI-GFP + Ta-211 AaBbDd C.sub.6C.sub.7C.sub.8C.sub.9C.sub.10 > T.sub.6T.sub.7T.sub.8T.sub.9A.sub.10
C.sub.6C.sub.7C.sub.8C.sub.9C.sub.10 > T.sub.6T.sub.7T.sub.8T.sub.9A.sub.10
C.sub.6C.sub.7C.sub.8C.sub.9C.sub.10 > T.sub.6T.sub.7T.sub.8T.sub.9A.sub.10 NO pUBI-A3A Ta-212 AABbDd WT
C.sub.6C.sub.7C.sub.8C.sub.9C.sub.10 > T.sub.6T.sub.7T.sub.8T.sub.9T.sub.10
C.sub.6C.sub.7C.sub.8C.sub.9C.sub.10 > T.sub.6T.sub.7T.sub.8T.sub.9T.sub.10 YES Ta-213 AaBbDd
C.sub.6C.sub.7C.sub.8C.sub.9C.sub.10 > T.sub.6T.sub.7T.sub.8T.sub.9A.sub.10 C.sub.6C.sub.7C.sub.8 > T.sub.6T.sub.7T.sub.8 C.sub.6C.sub.7C.sub.8 > T.sub.6T.sub.7T.sub.8 NO Shannong 116 pmTaGRF4- Ta-201 AaBbDd C.sub.6C.sub.7C.sub.8 > T.sub.6T.sub.7T.sub.8 C.sub.6C.sub.7C.sub.8 > T.sub.6T.sub.7T.sub.8;
C.sub.6C.sub.7C.sub.8C.sub.9C.sub.10 > T.sub.6T.sub.7T.sub.8T.sub.9T.sub.10 YES TaGIF1 + pUBI-A3A Ta-202 aabbdd CCCCCGCCGCAT > CCCCCGCCGCAT > C.sub.6C.sub.7C.sub.8C.sub.9 > T.sub.6T.sub.7T.sub.8T.sub.9 YES TTTTGCCGCTC ATTCCCCCACGA Ta-203 AaBbDd C.sub.6C.sub.7C.sub.8C.sub.9C.sub.10 > T.sub.6T.sub.7T.sub.8T.sub.9T.sub.10 C.sub.6C.sub.7C.sub.8C.sub.9C.sub.10 > T.sub.6T.sub.7T.sub.8T.sub.9T.sub.10
C.sub.6C.sub.7C.sub.8C.sub.9 > T.sub.6T.sub.7T.sub.8T.sub.9 NO Ta-204 AAbbddd WT C.sub.6C.sub.7C.sub.8C.sub.9 > T.sub.6T.sub.7T.sub.8T.sub.9 C.sub.6C.sub.7C.sub.8C.sub.9 > T.sub.6T.sub.7T.sub.8T.sub.9 NO Ta-205 AABbDD WT Del GTCCCCCGCCGCAT WT NO GATCGGCACGG Ta-206 AaBbDd C.sub.6C.sub.7C.sub.8C.sub.9C.sub.10 > T.sub.6T.sub.7T.sub.8T.sub.9A.sub.10 GGCCAGGTCCCCCGCCG C.sub.6C.sub.7C.sub.8C.sub.9C.sub.10 > T.sub.6T.sub.7T.sub.8T.sub.9A.sub.10 NO CATGAT Ta-207 AaBbDd C.sub.6C.sub.7C.sub.8C.sub.9C.sub.10 > T.sub.6T.sub.7T.sub.8T.sub.9A.sub.10 C.sub.6C.sub.7C.sub.8C.sub.9C.sub.10 > T.sub.6T.sub.7T.sub.8T.sub.9A.sub.10 C.sub.6C.sub.7C.sub.8C.sub.9C.sub.10 > T.sub.6T.sub.7T.sub.8T.sub.9A.sub.10 NO Jing 411 pUBI-GFP + pUBI-A3A Ta-301 AabbDd C.sub.6C.sub.7C.sub.8 > T.sub.6T.sub.7T.sub.8 C.sub.6C.sub.7C.sub.8 > T.sub.6T.sub.7T.sub.8 C.sub.6C.sub.7C.sub.8C.sub.9C.sub.10 > T.sub.6T.sub.7T.sub.8T.sub.9T.sub.10 NO C.sub.6C.sub.7 > T.sub.6T.sub.7 Ta-302 AaBbDd
C.sub.6C.sub.7C.sub.8C.sub.9C.sub.10 > T.sub.6T.sub.7T.sub.8T.sub.9A.sub.10 C.sub.6C.sub.7C.sub.8C.sub.9 > T.sub.6T.sub.7T.sub.8T.sub.9 C.sub.6C.sub.7C.sub.8 > T.sub.6T.sub.7T.sub.8 NO pmTaGRF4- Ta-401 aaBbDd C.sub.6C.sub.7C.sub.8 > T.sub.6T.sub.7T.sub.8 C.sub.6C.sub.7C.sub.8 > T.sub.6T.sub.7T.sub.8
C.sub.6C.sub.7C.sub.8C.sub.9C.sub.10 > T.sub.6T.sub.7T.sub.8T.sub.9T.sub.10 YES TaGIF1 + pUBI-A3A Ta-402 aaBbDd C.sub.6C.sub.7C.sub.8 > T.sub.6T.sub.7T.sub.8 C.sub.6C.sub.7C.sub.8 > T.sub.6T.sub.7T.sub.8
C.sub.6C.sub.7C.sub.8C.sub.9C.sub.10 > T.sub.6T.sub.7T.sub.8T.sub.9T.sub.10 YES Ta-403 aabbdd
C.sub.6C.sub.7C.sub.8C.sub.9C.sub.10 > T.sub.6T.sub.7T.sub.8T.sub.9T.sub.10; C.sub.6C.sub.7C.sub.8C.sub.9 > T.sub.6T.sub.7T.sub.8T.sub.9; A.sub.7C.sub.6C.sub.7C.sub.8C.sub.9 > G.sub.7T.sub.6T.sub.7T.sub.9 NO DEL CCCCCGCCGCAT C.sub.9 > G.sub.9 C.sub.6C.sub.7 > T.sub.6T.sub.7 Ta-404 aaBbDd C.sub.6C.sub.7C.sub.8 > T.sub.6T.sub.7T.sub.8 C.sub.6C.sub.7C.sub.8 > T.sub.6T.sub.7T.sub.8 C.sub.6C.sub.7C.sub.8C.sub.9C.sub.10 > T.sub.6T.sub.7T.sub.8T.sub.9T.sub.10 NO Ta-405 AaBbDd C.sub.6C.sub.7C.sub.8 > T.sub.6T.sub.7T.sub.8 C.sub.6C.sub.7C.sub.8 > T.sub.6T.sub.7T.sub.8 C.sub.6C.sub.7C.sub.8C.sub.9C.sub.10 > T.sub.6T.sub.7T.sub.8T.sub.9T.sub.10 NO Zhoumai 27 pUBI-GFP + pUBI-A3A Ta-431 AaBbDd C.sub.6C.sub.7 > T.sub.6T.sub.7 C.sub.6C.sub.7C.sub.8C.sub.9 > T.sub.6T.sub.7T.sub.8T.sub.9 C.sub.6C.sub.7C.sub.8C.sub.9 > T.sub.6T.sub.7T.sub.8T.sub.9 NO Ta-432 AaBbdd C.sub.6C.sub.7C.sub.8C.sub.9 > T.sub.6T.sub.7T.sub.8T.sub.9 C.sub.6C.sub.7C.sub.8C.sub.9 > T.sub.6T.sub.7T.sub.8T.sub.9 C.sub.6C.sub.7C.sub.8C.sub.9 > T.sub.6T.sub.7T.sub.8T.sub.9 NO C.sub.6C.sub.7C.sub.8C.sub.9C.sub.10 > T.sub.6T.sub.7T.sub.8T.sub.9T.sub.10 Ta-433 AaBbDd C.sub.6C.sub.7C.sub.8 > T.sub.6T.sub.7T.sub.8 C.sub.6C.sub.7C.sub.8C.sub.9 > T.sub.6T.sub.7T.sub.8T.sub.9 C.sub.6C.sub.7C.sub.8 > T.sub.6T.sub.7T.sub.8 YES Ta-434 AaBbDd C.sub.6C.sub.7 > T.sub.6T.sub.7 C.sub.6C.sub.7C.sub.8 > T.sub.6T.sub.7T.sub.8 C.sub.6C.sub.7C.sub.8 > T.sub.6T.sub.7T.sub.8 NO pmTaGRF4- Ta-450 AaBbdd C.sub.6C.sub.7C.sub.8 > T.sub.6T.sub.7T.sub.8 C.sub.6C.sub.7C.sub.8C.sub.9C.sub.10 > T.sub.6T.sub.7T.sub.8T.sub.9T.sub.10 C.sub.6C.sub.7C.sub.8C.sub.9C.sub.10 > T.sub.6T.sub.7T.sub.8T.sub.9T.sub.10 NO TaGIF1 + pUBI-A3A Ta-451 aaBBDd C.sub.6C.sub.7C.sub.8 > T.sub.6T.sub.7T.sub.8 WT WT YES C.sub.-5 > T.sub.-5 Ta-452 Aabbdd C.sub.6C.sub.7C.sub.8C.sub.9 > T.sub.6T.sub.7T.sub.8T.sub.9 C.sub.6C.sub.7C.sub.8C.sub.9C.sub.10 > T.sub.6T.sub.7T.sub.8T.sub.9T.sub.10; C.sub.6C.sub.7C.sub.8C.sub.9C.sub.10 > T.sub.6T.sub.7T.sub.8T.sub.9T.sub.10 YES C.sub.6C.sub.7C.sub.8C.sub.9 >

T.sub.6T.sub.8T.sub.9 > T.sub.6T.sub.7T.sub.8T.sub.9 Ta-453 aabdd
 C.sub.6C.sub.7C.sub.8C.sub.9C.sub.10 > T.sub.6T.sub.7T.sub.8T.sub.9T.sub.10
 C.sub.6C.sub.7C.sub.8C.sub.9C.sub.10 > T.sub.6T.sub.7T.sub.8T.sub.9T.sub.10
 C.sub.6C.sub.7C.sub.8C.sub.9C.sub.10 > T.sub.6T.sub.7T.sub.8T.sub.9T.sub.10 YES C.sub.6C.sub.7C.sub.8 >
 T.sub.6T.sub.7T.sub.8 C.sub.6C.sub.7C.sub.8 > T.sub.6T.sub.7T.sub.8 Ta-454 aabbDd
 C.sub.6C.sub.7C.sub.8C.sub.9C.sub.10 > T.sub.6T.sub.7T.sub.8T.sub.9T.sub.10
 C.sub.6C.sub.7C.sub.8C.sub.9C.sub.10 > T.sub.6T.sub.7T.sub.8T.sub.9T.sub.10; C.sub.10 > T.sub.10 NO
 C.sub.6C.sub.7C.sub.8 > T.sub.6T.sub.7T.sub.8 C.sub.6C.sub.7C.sub.8C.sub.9 > T.sub.6T.sub.7T.sub.8T.sub.9 Ta-455
 AabbDd C.sub.6C.sub.7 > T.sub.6T.sub.7 C.sub.6C.sub.7C.sub.8C.sub.9C.sub.10 >
 T.sub.6T.sub.7T.sub.8T.sub.9T.sub.10; C.sub.6C.sub.7C.sub.8 > T.sub.6T.sub.7T.sub.8 NO
 C.sub.6C.sub.7C.sub.8C.sub.9 > T.sub.6T.sub.7T.sub.8T.sub.9 Ta-456 AaBbDd C.sub.6C.sub.7C.sub.8 >
 T.sub.6T.sub.7T.sub.8 C.sub.6C.sub.7C.sub.8C.sub.9 > T.sub.6T.sub.7T.sub.8T.sub.9 C.sub.6C.sub.7 > T.sub.6T.sub.7
 YES Ta-457 AaBbDd C.sub.6C.sub.7C.sub.8 > T.sub.6T.sub.7T.sub.8 C.sub.6C.sub.7C.sub.8C.sub.9 >
 T.sub.6T.sub.7T.sub.8T.sub.9 C.sub.6C.sub.7 > T.sub.6T.sub.7 YES Zhoumai 28 pmTaGRF4- Ta-411 AaBbDd
 C.sub.6C.sub.7C.sub.8C.sub.9C.sub.10 > T.sub.6T.sub.7T.sub.8T.sub.9T.sub.10
 C.sub.6C.sub.7C.sub.8C.sub.9C.sub.10 > T.sub.6T.sub.7T.sub.8T.sub.9T.sub.10 C.sub.6C.sub.7C.sub.8 >
 T.sub.6T.sub.7T.sub.8 YES TaGIF1 + pUBI-A3A Ta-412 aabdd C.sub.6C.sub.7C.sub.8C.sub.9C.sub.10 >
 T.sub.6T.sub.7T.sub.8T.sub.9T.sub.10 C.sub.6C.sub.7C.sub.8 > T.sub.6T.sub.7T.sub.8
 C.sub.6C.sub.7C.sub.8C.sub.9C.sub.10 > T.sub.6T.sub.7T.sub.8T.sub.9T.sub.10 NO Ta-413 AaBbDd
 C.sub.6C.sub.7C.sub.8 > T.sub.6T.sub.7T.sub.8 C.sub.6C.sub.7C.sub.8 > T.sub.6T.sub.7T.sub.8
 C.sub.6C.sub.7C.sub.8C.sub.9C.sub.10 > T.sub.6T.sub.7T.sub.8T.sub.9T.sub.10 YES Ta-414 aaBBDD
 C.sub.6C.sub.7C.sub.8 > T.sub.6T.sub.7T.sub.8 WT WT NO Ta-415 aabdd C.sub.6C.sub.7C.sub.8C.sub.9C.sub.10 >
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 YES Xiaoyan 54 pmTaGRF4- Ta-501 AaBbdd C.sub.6C.sub.7C.sub.8C.sub.9C.sub.10 >
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 T.sub.6T.sub.7T.sub.8T.sub.9 C.sub.6C.sub.7C.sub.8 > T.sub.6T.sub.7T.sub.8 C.sub.6C.sub.7C.sub.8C.sub.9C.sub.10
 > T.sub.6T.sub.7T.sub.8T.sub.9A.sub.10 NO Aa/aa, Bb/bb, and Dd/dd represent heterozygous/homozygous mutations
 in TaALS-1A, TaALS-1B, and TaALS-1D, respectively.

(151) TABLE-US-00016 Sequences SEQ ID NO: 1 ZmWUS amino acid sequence
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18 GmGRF6-GmGIF1

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TEEEGS

Claims

1. A method for improving plant cell regeneration efficiency, the method comprising: (a) introducing into a cell of the plant i) an expression construct comprising a coding nucleic acid sequence of WUS, an expression construct comprising a coding nucleic acid sequence of BBM, and an expression construct comprising a nucleic acid sequence of SERK, wherein the WUS comprises the amino acid sequence shown in SEQ ID NO: 1, the BBM comprises the amino acid sequence shown in SEQ ID NO: 2, or the SERK comprises the amino acid sequence shown in SEQ ID NO: 3; and/or ii) an expression construct comprising a coding nucleic acid sequence of GRF and an expression construct comprising a coding nucleic acid sequence of GIF, wherein the GRF comprises the amino acid sequence of SEQ ID NO: 7 and the GIF1 comprises the amino acid sequence shown in SEQ ID NO: 11; and/or iii) an expression construct comprising the nucleotide sequence of SEQ ID NO: 22; (b) regenerating an intact plant from the plant cell.
2. A method of transforming at least one exogenous nucleic acid sequence of interest into a plant, the method comprising: (a) introducing into a cell of the plant i) an expression construct comprising a coding nucleic acid sequence of WUS, an expression construct comprising a coding nucleic acid sequence of BBM and an expression construct comprising a coding nucleic acid sequence of SERK, wherein the WUS comprises the amino acid sequence shown in SEQ ID NO: 1, the BBM comprises the amino acid sequence shown in SEQ ID NO: 2, or the SERK comprises the amino acid sequence shown in SEQ ID NO: 3; and/or ii) an expression construct comprising a coding nucleic acid sequence of GRF and an expression construct comprising a coding nucleic acid sequence of GIF, wherein the GRF comprises the amino acid sequence of SEQ ID NO: 7, and the GIF1 comprises the amino acid sequence shown in SEQ ID NO: 11; and/or iii) an expression construct comprising the nucleotide sequence of SEQ ID NO: 22; (b) introducing at least one expression construct comprising the at least one exogenous nucleic acid sequence of interest into the plant cell; and (c) regenerating an intact plant from the plant cell.
3. A method for gene editing in a plant, the method comprising: (a) introducing into a cell of the plant i) an expression construct comprising a coding nucleic acid sequence of WUS, an expression construct comprising a coding nucleic acid sequence of BBM and an expression construct comprising a coding nucleic acid sequence of SERK, wherein the WUS comprises the amino acid sequence shown in SEQ ID NO: 1, the BBM comprises the amino acid sequence shown in SEQ ID NO: 2, or the SERK comprises the amino acid sequence shown in SEQ ID NO: 3; and/or ii) an expression construct comprising a coding nucleic acid sequence of GRF and an expression construct comprising a coding nucleic acid sequence of GIF, wherein the GRF comprises the amino acid sequence of SEQ ID NO: 7, and the GIF1 comprises the amino acid sequence shown in SEQ ID NO: 11; and/or iii) an expression construct comprising the nucleotide sequence of SEQ ID NO: 22; (b) introducing at least one expression construct comprising the at least one exogenous nucleic acid sequence of interest into the plant cell, wherein the at least one exogenous nucleic acid sequence of interest encodes a component of a gene editing system; or introducing at least one component of the gene editing system into the plant cell; and (c) regenerating an intact plant from the plant cell.
4. The method according to any one of claims 1-3, wherein the coding nucleic acid sequence of WUS, the coding nucleic acid sequence of BBM and the coding nucleic acid sequence of SERK are placed in a same expression construct.
5. The method according to claim 4, wherein the coding nucleic acid sequence of WUS, the coding nucleic acid sequence of BBM, the coding nucleic acid sequence of SERK, and the at least one exogenous nucleic acid sequence of interest, if present, are placed in a same expression construct.
6. The method according to any one of claims 1-3, wherein the coding nucleic acid sequence of GRF and the coding nucleic acid sequence of GIF are placed in a same expression construct.
7. The method according to claim 6, wherein the coding nucleic acid sequence of GRF the coding nucleic acid sequence of GIF and the at least one exogenous nucleic acid sequence of interest, if present, are placed in a same expression construct.
8. The method according to any one of claims 1-3, wherein the GRF comprises a mutated miRNA binding site, so as not to be regulated by the miRNA, wherein the miRNA is miR396.
9. The method according to any one of claims 1-3, wherein the GRF is fused to the GIF by a linker, and further

wherein the linker comprises an amino acid sequence comprising SEQ ID NO: 12.

10. The method according to claim 3, wherein the gene editing system is selected from the group consisting of CRISPR system, TALEN, meganuclease and zinc finger nuclease.

11. The method according to claim 10, wherein the gene editing system is a base editing system.

12. The method according to claim 3, wherein the gene editing system is a base editor comprising an amino acid sequence comprising SEQ ID NO: 14.
